

MALAYSIAN GUIDELINES

In the treatment of

SEXUALLY TRANSMITTED INFECTIONS



FOURTH EDITION
2015



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SEXUALLY TRANSMITTED INFECTIONS

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CONTRIBUTING AUTHORS

Technical Advisors	Datuk Dr Roshidah bt Baba, FRCP Head of Dermatological Services MOH Senior Consultant Dermatologist Hospital Melaka
	Dr Sha'ari bin Ngadiman, MPH Public Health Physician Head of HIV/AIDS Sector, Ministry of Health Malaysia
Chairman	Dr Najeeb Ahmad bin Mohd Safdar, FRCP Consultant Dermatologist Hospital Tuanku Jaafar Seremban
Co-chairman	Dr Raja Iskandar bin Raja Azwa, MRCP Assoc Professor (Infectious Disease Physician) University Malaya Medical Centre

Members

Dr Zubaidah bt Abdul Wahab, MPath (Med Microbiology) Senior Consultant Clinical Microbiologist Pathology Department Hospital Sungai Buloh, Selangor	Dr. Akbal Kaur, Dip.Derm Senior Medical Officer (Rtd) Hospital Kuala Lumpur Kuala Lumpur
Dr. Sabeera Begum, MRCP(Paed) Consultant Paediatrician Institute of Paediatrics Hospital Kuala Lumpur, Kuala Lumpur	Dr Felix Yap Boon Bin, MRCP Dermatologist, Universiti Tuanku Abdul Rahman
Assoc Prof Dr Khong Su Yen, MRCOG, FRANZCOG Obstetrician & Gynaecologist University Malaya Medical Centre Kuala Lumpur	Dr Noorlaily bt Mohd Noor, MRCP Dermatologist Hospital Kuala Lumpur
Dr Nor Azah bt Mohd Nawi, MMed (Fam Med) Family Medicine Specialist Klinik Kesihatan Bakar Arang	Dr Nazrila Hairizan bt Nasir, MMed (Fam Med) Family Medicine Specialist Pandamaran Health Clinic Klang, Selangor

Dr Ahmad Zukri bin Mat Ail, MMed (Fam Med)
Family Medicine Specialist
Ketereh Health Clinic, Kota Bharu, Kelantan

Dr Ilias Adam Yee, LRCPI / SI,
Executive Director,
Malaysian AIDS Council.

Late Dr Doshi HK (Dip. Derm, Dip. GUM)
Private Clinic(Shriji Skin & STI Clinic)
Sekinchan, Selangor

Dr Marniza bt Abdul Aziz (MPath (Med Microbiology)
Clinical Microbiologist
Pathology Department
Hospital Sg Buloh, Selangor

Dr Tuan Suhaila bt Tuan Soh (MPath (Med Microbiology)
Clinical Microbiologist
Pathology Department
Hospital Sg Buloh , Selangor

Dr. Mohd Nasir bin Abd Aziz, MPH
Public Health Physician
Senior Principal Assistant Director
HIV/STI Unit
Wilayah Persekutuan State Health Department, WP Kuala Lumpur

Dr Asiah bt Ayob, MPH
Public Health Physician
Senior Principal Assistant Director
Communicable Disease Unit (Surveillance)
Perak State Health Department, Ipoh

Puan Siti Jamiza bt Mohd Saim,
Pharmacist
Hospital Kuala Lumpur

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FOREWORD

I would like to congratulate the committee for producing the Fourth Edition of The Malaysian Guidelines in The Treatment of Sexually Transmitted Infections.

Given the social, demographic and migratory trends within the last few decades around the region and the world, there will be segments of populations to be at higher risk for sexually transmitted infections (STIs). Globally, there is a rising trend of non-curable viral STIs. Besides serious delayed consequences (from STIs) such as infertility, ectopic pregnancy, cervical cancer and the untimely deaths of infants and adults, the economic burden and loss of productivity to individuals and nations as part of social sequelae to STIs will always be an issue.

The Ministry of Health of Malaysia has a long history of providing STI related health services to the general population. Strategies have been adapted from around the world to suit local situations and culture – “*Global evidence, localize decision*”. In the last decade, the National HIV / STI Prevention and Control Programme has engaged public and private sectors in the integration of the key elements of reproductive health care services.

STI prevention and control programme encompasses adolescent and women’s health, family planning, safe motherhood, immunization, child survival and HIV prevention. An effective partnership within the various divisions of the Ministry of Health (MOH) will determine the success of the programme.

Effective prevention and care can be achieved by use of a combination of responses. Services for prevention and care for people with STIs should embrace a comprehensive public health package. This includes promotion of safer sexual behavior, early health-care-seeking behavior, and a comprehensive approach to case management.

MOH will continue to put emphasis on a streamlined STI surveillance system. A two yearly Integrated Bio-Behavioural Surveillance Survey (IBBS) since 2004 helps to enhance the existing STI surveillance system. This approach will provide STI programmes with new information on appropriate interventions that are evidence based.

I believe that this guideline will be of tremendous use to all healthcare personnel involved in the managements of patients with STIs.



DATUK DR. NOOR HISHAM BIN ABDULLAH
DIRECTOR GENERAL OF HEALTH MALAYSIA

PREFACE

I wish to take this opportunity to thank and congratulate the secretariat and the working committee for painstakingly taking their time off to revise *The Malaysian Guidelines in the Treatment of Sexually Transmitted Infections Year 2008*. Accurate diagnosis and effective treatment complemented with a comprehensive package on STIs are of utmost importance in managing patients with STIs. Thus, I believe it is timely that this revision is done using an inclusive and broad consultative process over the past year within the framework of the Ministry of Health and our external partners.

The World Health Organization estimates that more than 340 million new cases of sexually transmitted bacterial and protozoal infections occur throughout the world every year. It is estimated that untreated early syphilis will result in a stillbirth rate of 25%, and be responsible for 14% of neonatal deaths, and an overall perinatal mortality of about 40%. Untreated Gonococcal and Chlamydial infections in women will result in pelvic inflammatory disease in up to 40% of cases, and 25% of them will result in infertility.

Consistent availability of appropriate treatment is essential for a successful STI control programmes. Malaysia is fortunate, unlike in some parts of the world where there is limited or no access to appropriate diagnostics. Though our infrastructure to help us treating patients with STIs is considered good, we still grapple with situations whereby patients are reluctant to seek treatment for fear of stigma and discrimination. Thus, there are areas in which we need to upscale our services, for example, new point-of-care rapid tests to enable screening for syphilis at the peripheral health posts, and thus provision of treatment without delay in this country is definitely warranted.

At the same time, the Ministry of Health does not overlook the need of other tests for example Nucleic Acid Amplification tests which we hope to be readily available in most settings, whereas in some, Polymerase Chain Reaction (PCR) definitely helps. Over the last few years, some medicines, for example, such as Azithromycin and Acyclovir have become more readily available in most primary health care settings. There are other areas for example that I am indeed pleased to note that the Ministry of Health have implemented for example the Ministry has embarked preventive vaccination against the Oncogenic Human Papilloma Virus for schoolgirls over the last 3 years and has become a standard package for the immunization programme.

Finally, allow me to thank all those who helped to make this revision possible, and to those who helped to elaborate and provided information and critically review the drafts. Particular thanks go to colleagues in the Universities, our very own Dermatologists, Public Health Physicians and Family Medicine Specialists who made it possible for the consultations to take place. It is my sincere hope that this guideline will help us attain the goals and targets as addressed in the *Global Strategy for the prevention and control of sexually transmitted infections 2006-2015* as outlined by the WHO. Thank you.

Dr Najeeb Ahmad bin Mohd Safdar
Chairman 2014/2015

THE CHALLENGES IN THE PREVENTION AND CONTROL OF SEXUALLY TRANSMITTED INFECTIONS (STIs) IN MALAYSIA

At the Millennium Summit in 2000, 189 countries around the world committed to the attainment of a set of Millennium Development Goals (MDG). Goal number 6 calls for halting and reversing the spread of HIV by 2015. While some countries are still trying to achieve these MDG goals, Malaysia has so far invested significant time and resources which have resulted in significant reduction in the spread of HIV in the country. In 2006, the HIV notification rate was 21.9 per 100,000 population and it dropped to 11.7 per 100,000 population in 2014. Meanwhile, trend of the 3 notifiable STIs (Chancroid, Gonorrhea and Syphilis) in Malaysia has remained stable; in 2006 the prevalence of Chancroid was 0 per 100,000, 1.9 for gonorrhea and 3.06 for syphilis. In 2014, the incidence of Chancroid remained 0.00, Gonorrhea 5.88, and Syphilis 5.61 per 100,000 population respectively. Several studies in Malaysia have shown the prevalence of gonorrhea and syphilis to be between 5-10%.

Much needs to be done to control the spread of STIs in Malaysia and around the region. For example, despite the fact that the government has a national policy for antenatal screening to prevent congenital syphilis and HIV, sporadic cases of congenital syphilis (and HIV) occur from time to time especially among the immigrant populations.

The Ministry of Health has to tackle the increasing challenges in the prevention and control of STI by ensuring continuation of sustainable programmes and activities linked to HIV prevention.

Some key challenges faced by health care providers in Malaysia in controlling STIs are discussed below:

1.0 Targeted interventions for most-at-risk populations (MARPS)

The Ministry of Health Malaysia (MOH) has played a key role for targeted interventions for MARPS through various channels, programmes and activities for many years. Some of these activities and programmes were implemented through networking, principally with the Malaysian AIDS Council (MAC) and other partner non government organization (NGO). Over the last 5 years, various new networking linkages have been formed i.e. the Global Fund for TB, AIDS and Malaria (GFTAM) and European Alliance which are based in Geneva and London respectively.

Ministry of Health (MOH) emphasizes on health care delivery for all segments of the population, including MARPS. Much of the emphasis has been placed on up scaling the delivery of the services which include increasing the coverage, providing quality services and scope of interventions. Some of these projects which have been in place for many years, were initially established to be more accessible to all segments of the population and over the years “rebranded”, staff retrained, and targeted to MARPS and other key affected populations.

“STI friendly clinics” which are more accessible and acceptable to MARPs, were established by the MOH through linkages with our partner organisations. In some cases, the MOH has also supported the establishment of drop-in-centres which provide a safe haven for these key affected populations. This option was undertaken to enhance the collaboration between the government physicians and the NGOs. The practice of linking MARPS and other key affected populations through partnerships is thought to be more sustainable.

Family physicians have been treating MARPS for many years. Through these specialized treatment and care, free distribution of condoms and other interventions that were incorporated, interruption of transmission of HIV and STIs through sex workers and their clients would be achieved.

MOH hopes that more of such STI friendly clinics would be established in the future. Klinik Kesihatan Pandamaran in Selangor, Klinik Kesihatan Batu in Kuala Lumpur and Georgetown Penang have all proven that such establishments and partnerships work.

2.0 Elimination of vertical transmission of congenital syphilis

Malaysia's long standing prevention of mother to child transmission (PMTCT) for syphilis has made impressive gains. The government mandated a national policy to screen antenatal mothers for syphilis more than 30 years ago.

Currently, patients with VDRL/RPR and TPHA/TPPA positive are assessed and treated appropriately. Referrals and a good follow-up system are in place to minimize loss of pregnant women with syphilis to follow-up.

It is estimated that some stillbirths could be due to syphilis, yet we do not have enough data to support this. Future research is needed to address syphilis as a cause of stillbirths. Mothers who are found to be TPHA positive need to be closely monitored. It would be ideal if a second evaluation (VDRL) is performed after 28 weeks of pregnancy.

3.0 Quality assurance and the STI training for key personnel

Malaysia has a good public health system with various programmes and activities incorporated into our primary care facilities over the last few decades to facilitate the delivery of services. The quality of delivery of services into various programmes and activities should continue to be improved

Training of Medical Laboratory Technologists (MLT) who currently form the backbone of the laboratories in the Primary Health Care (PHC) needs to be constantly reviewed and enhanced. It is therefore critical that the MOH set up quality assurance schemes to ensure the quality of tests performed in the PHCs are not compromised, and that the health workers are proficient in performing these tests.

There is an urgent need for other newer tests for various STIs to be made available either in hospitals or primary health clinics e.g. chlamydia PCR or immune-fluorescence, HSV and GC cultures and better point of care (rapid) tests with increased sensitivities.

4.0 Access to services for the adolescents and other special groups

Clinics specializing in STI treatment are helpful in providing primary care in urban settings for specific groups (e.g. sex workers, migrant workers, truckers, PWID and their sexual partners). With the expertise available (e.g. Family Medicine Specialists with special interest in STIs and HIV/AIDS, sexual reproductive health, adolescent health clinics), these clinics could provide referral services to hospital outpatient departments and private practitioners while functioning as training centres to train other healthcare providers.

Adolescents for example, often lack information about STI services, and even if they know about these services, they are often reluctant to seek help for diagnosis and treatment, and are often embarrassed and worried about social stigmatization. There needs to be initiatives to make health services more adolescent-friendly and more responsive to their particular needs.

Points for consideration for the implementation of STI programmes

(a) Measuring effectiveness of the STI programmes

Monitoring and evaluation of STI programmes and activities should be an on-going process. The establishment and improvement of existing monitoring systems will be needed, with regular evaluation of programme interventions.

The STI prevention and control programmes have been incooperated into HIV/AIDS programmes for many years. Much of the emphasis so far have been placed on evaluating solely on HIV/AIDS programmes. Therefore, concerted efforts are needed to measure the effectiveness of individual STI programmes e.g. prevalence of syphilis among antenatal mothers, stillbirths deaths due to syphilis in pregnancy and congenital syphilis, prevalence of chlamydia among teenagers and schoolchildren just to name a few.

(b) Strategies for replication

Best practices projects should allow interested parties to replicate projects which are deemed “successful”. Efforts should be made to replicate these successful projects, and it should be done in a way that will allow local groups to adopt and adapt the overall strategies to suit their own situations.

STI prevention and control projects in Malaysia need the involvement of a lot of key stakeholders ie NGOs, local health authorities, the Ministry of Health etc. Expansion or replication of sexual health related projects should allow mutual understanding amongst the key stakeholders to build up future partnerships. These key stakeholders need to be convinced the value of such projects (and partnerships) in the long run.

(c) Efficient management and good governance.

In any organization, an efficient management and good governance plays a key role in retaining staff and making it worthwhile for the employees to continue staying on. For the fund providers, an efficient organization which is deemed efficient and successful will make it worthwhile to continue providing funds and keep the projects sustainable. Among others, the key elements that the fund providers would seek in continuing giving funds are “good budgetary control”, “direct supervision of the activities”, “high returns for investments”, “impact of the programmes” etc.

Managers of government agencies and community based organisations need to understand that financial resources need to be in place in order to ensure sustainability of projects. Thus, for programmes / activities to be sustainable, it is crucial that those who hold to key positions in any organization (government, private or community based) should be able to impress the fund providers that the money being channeled is well spent.

(d) Ensure constant advocacy for a successful STI health initiative

Key programme managers should ensure and seek to mobilise commitment and advocacy at local, national or international levels. They need to understand for example, the value of linking congenital syphilis elimination to other antenatal health services, PMTCT of HIV programmes and other STI prevention and control programmes. A sustainable programme is only possible if there is commitment at all a level. Policies, strategies, and programmes need to be reviewed periodically to ensure that the problems are addressed and strategies implemented.

GENERAL SCREENING

Sexually transmitted infections are recognized worldwide as important causes of morbidity in both maternal and child health. They also pose a major threat to reproductive health.

Sexually transmitted bacterial infections can be cured by accurate diagnosis and prompt and effective treatment rendering the patient non-infectious. Complications and further transmission can be prevented by contact tracing and effective therapy of infected contacts.

Currently available therapy for sexually transmitted viral diseases does not ensure a cure and exerts a much less, if any, effect on transmission. All the viruses remain latent for long periods. Sexually transmitted viral infections usually follow a clinical course of remissions and relapses, and therapy does not always render the patient non-infectious. Asymptomatic infected persons can be infectious to their partners. Education, counseling and positive behavioral change are the main ways to prevent transmission of sexually transmitted viral infections, including human immunodeficiency viral (HIV) infection.

This guideline is issued with the hope that at least the sexually transmitted bacterial infections are managed effectively and every opportunity is taken to educate and counsel patients in order to promote a lifestyle devoid of careless sexual activity and condom usage if the lifestyle is not adhered to. Prevention and control of STIs is of utmost importance in this HIV era since it is well-known that the presence of a STI, whether ulcerative or non-ulcerative, may enhance both the acquisition and transmission of HIV.

TAKING A SEXUAL HEALTH HISTORY**1. General rules**

- Ensure privacy and confidentiality
- Adopt polite, friendly, non-judgmental attitude
- Ask open-ended questions to initiate dialogue
- Phrase questions to minimize opportunity to mislead
- Proceed with close-ended questions once the patient is comfortable
- more questions may need to be asked during the examination
- Do not show annoyance if the patient is not cooperative

2. Presenting complaint

- Obtain detailed history of presenting symptom(s)
- Enquire about the presence of other symptoms that are common

- Females:
 - Vaginal discharge
 - Genital ulceration
 - Dysuria, frequency
 - Vulval itching
 - Abnormal growth or mass in genital area
 - Dyspareunia (pain during sexual intercourse)
 - Post-coital bleeding
 - Intermenstrual bleeding
 - Perianal pain
 - Anal discharge
 - Inguinal lymphadenopathy
 - Lower abdominal pain
- Males:
 - Dysuria, frequency
 - Urethral discharge
 - Genital ulceration
 - Abnormal growth or mass in genital area
 - Acute scrotal swelling, pain
 - Perianal pain
 - Anal discharge
 - Inguinal lymphadenopathy

3. Sexual history

Number of partners in the past three months (for each partner; to ask and document the following):

- Date of last sexual intercourse
- Relationship to partner (s) (spouse, regular non-spouse, casual)
- Gender of Partner
- Duration of sexual relationship (for regular partners)
- Type of exposure (oral, vaginal, anal)& role (insertive, receptive)
- Use of condoms
- Problems or symptoms in the partner(s)
- Last episode of unprotected anal/vaginal intercourse

4. Other illnesses

5. Medications

6. Drug allergies

7. Gynaecological and Obstetric history

- Date of last menstrual period
- Regular / Irregular periods
- Missed or overdue period
- Pain during periods
- Excessive bleeding
- Post-coital bleeding
- Number of pregnancies and outcomes including abortions
- Mode of delivery
- Contraception

8. Risk factors

a) Occupation

- Sex worker (male and female)
- Seamen
- Workers in the tourist industry
- Transport workers
- Migrant workers

b) Travel history

c) Unprotected sexual encounters

d) Multiple sexual partners

e) Previous history of STIs

- Presenting symptoms and treatment
- Most recent HIV test and results
- Most recent STI screening and results

f) History of injections or blood transfusions

g) Substance use: alcohol, drugs etc

h) Tattooing

i) History of buying or selling sex

j) History of Hepatitis B vaccination

GENERAL GUIDELINES

All patients requesting evaluation of STI should undergo the following as part of their routine health care:

- History
- Medical & behavioural risk assessment
- Laboratory investigations
- Diagnosis

- Curative or palliative therapy
- Counseling & education
- Management of present episode of STI
- Prevention of future episodes
- Reporting of a case*
- Sex partner identification, notification and evaluation
- Clinical follow up where appropriate

* Only Syphilis, Chancroid, Gonorrhoea & HIV infections are notifiable in Malaysia to date.

ROUTINE LABORATORY TESTS

ALL PATIENTS

BLOOD TEST	MICROSCOPY
HIV antibody test	Dark-field microscopy for <i>T. pallidum</i>
Syphilis serology (*both RPR and TPHA/TPPA)	HSV antigen detection for all genital ulcers
HBs Ag	
Anti HCV Ab	

Note: * For Syphilis screening perform TPHA/TPPA and proceed with RPR if positive

Male	Female
Urethral swab - Gram stain for pus cells and intracellular Gram-negative diplococci - Culture for <i>N. gonorrhoeae</i> - Antigen / NAAT test for <i>C. trachomatis</i> Sub-preputial swab - Gram stain for Candida yeast	Endocervical swabs - Gram stain for pus cells, intracellular Gram-negative diplococci - Culture for <i>N. gonorrhoeae</i> - Antigen / NAAT test for <i>C. trachomatis</i> - Pap smear Vaginal swabs - Wet mount from posterior fornix for <i>Trichomonas vaginalis</i> - Gram stain for pus cells, yeasts and clue cells - Candida culture - swab from lateral fornix

Indications for lumbar puncture in patients with Syphilis

- i. If neurosyphilis is suspected
- ii. For late latent syphilis and syphilis of unknown duration only in HIV-infected patients
- iii. For ocular syphilis

SYPHILIS

Aetiology

Treponema pallidum

Classification

Acquired	
Early Syphilis (<2 Years)	Late Syphilis (>2 Years)
• Primary Syphilis • Secondary Syphilis • Early Latent Syphilis	• Late Latent Syphilis • Tertiary Syphilis (Gumma) • Cardiovascular Syphilis • Neurosyphilis
Congenital	
Early (<2 Years Old)	Late (> 2 Years Old)

EARLY SYPHILIS

Early syphilis is defined as infection during the first 2 years and includes primary, secondary and early latent syphilis.

PRIMARY SYPHILIS

Incubation Period: 9-90 days

Presentation / findings

Primary syphilis presents classically as a solitary, non-tender, indurated and well-circumscribed ulcer (chancre) with regional lymphadenopathy. The chancre is commonly located in the anogenital region. However, chancres may be multiple, painful, purulent and extragenital (most frequently oral). It may present as syphilitic balanitis of Follman.

Any anogenital ulcer should be considered to be due to syphilis unless proven otherwise.

Diagnosis

- Dark-field microscopy for *T. pallidum* from ulcer base
- Direct immunofluorescent test (DFAT) for *T. pallidum*
- PCR for *T. pallidum*
- Non-specific serology with non-treponemal tests RPR - if negative, repeat at 1 month & 3 months
- Specific treponemal tests (TPHA / TPPA) are less sensitive than non-treponemal tests in early primary syphilis
- The syphilis EIA is reported to be more sensitive than other specific treponemal tests in early syphilis

SECONDARY SYPHILIS

Incubation Period: 6 weeks - 6 months

Presentation

The commonest presentation is a generalized non-irritating skin lesion involving the palms and soles with or without generalized lymphadenopathy. Condylomata lata, mucocutaneous lesions and patchy alopecia are seen less commonly. The rash is classically non-itchy but may be itchy, particularly in dark-skinned patients.

Secondary syphilis is a stage of bacteraemia and hence, patients can present with symptoms and/ or signs referable to any system in the body for example anterior uveitis, meningitis, cranial nerve palsies, hepatitis, splenomegaly, periosteitis and glomerulonephritis.

Diagnosis

- All serological tests for syphilis are expected to be positive in secondary syphilis
- RPR titres in untreated cases are often > 1:8 (RPR)
- If a non-treponemal test (RPR) is used for diagnosis, confirm a positive result with a specific treponemal test (TPHA / TPPA / EIA)
- If a specific treponemal test is used for diagnosis and is found to be positive, use the RPR test to determine disease activity, and to monitor response to therapy
- Dark field microscopy / DFAT from mucocutaneous lesions

EARLY LATENT SYPHILIS

Early Latent Syphilis is diagnosed by a positive serology without symptoms and signs in a person known to be sero-negative in the previous 2 years

TREATMENT

Treatment of Early Syphilis (*Primary, Secondary and Early Latent Syphilis*) and Late Syphilis (*Late Latent, Gumma, Cardiovascular and Neurosyphilis*)

Early Syphilis	Late Syphilis (excluding Neurosyphilis)
Recommended Regimen	
- Benzathine penicillin, 2.4 mega units I.M. in a single dose (<i>Grade A, Ib</i>) OR - Procaine penicillin G, 600,000 units I.M. daily for 10 days (<i>Grade B, III</i>)	- Benzathine penicillin, 2.4 mega units I.M. weekly X 3 weeks (3 doses) (<i>Grade B, III</i>) OR - Procaine penicillin G, 600,000 units I.M. daily for 17 days (<i>Grade B, III</i>)
Alternative Regimen (These may be required for those with penicillin allergy)	
- Ceftriaxone 500 mg I.M. daily for 10 days (<i>Grade B, I</i>) - <i>if no anaphylaxis to penicillin</i> OR - Doxycycline 100 mg b.d. P.O. for 14 days (<i>Grade B, III</i>); OR - Erythromycin stearate 500 mg q.i.d. P.O. for 14 days (<i>Grade B, III</i>); OR - Erythromycin ethyl succinate 800 mg q.i.d. P.O. x 14 days (<i>Grade B, III</i>); OR - Azithromycin 2 g single dose P.O. (<i>Grade B, II</i>) (<i>concerns regarding intrinsic macrolide resistance</i>)	- Doxycycline 100 mg b.d. P.O. for 28 days (<i>Grade C, IV</i>); OR - Erythromycin stearate 500 mg q.i.d. P.O. for 28 days (<i>Grade C, IV</i>); OR - Erythromycin ethyl succinate 800 mg q.i.d. P.O. x 28 days (<i>Grade C, IV</i>);

*Treatment for mother who detected positive for syphilis: penicillin regimen appropriate for the women's stage of syphilis.

Advice

Patients should be advised to abstain from sex until 1 week after they and their partner(s) have completed treatment.

Contact Tracing

All sexual partners should be examined, investigated and treated epidemiologically. For patients with primary syphilis, sexual partners within the past three months should be notified as the incubation period is up to 90 days. Partner notification may have to extend to 2 years for patients in secondary syphilis with clinical relapse or in early latent syphilis.

All patients should be offered patient and provider referral as a method of contacting any sexual partner. The method agreed upon with the patient should be clearly documented. Epidemiological treatment for asymptomatic contacts of early syphilis is recommended.

INCUBATING SYPHILIS / EPIDEMIOLOGICAL TREATMENT**Recommended Regimen**

- Benzathine penicillin 2.4 MU I.M. single dose (Grade B, III)

Alternative Regimen

- Doxycycline 100 mg b.d. P.O. for 14 days (Grade B, III); OR
- Azithromycin 1 gm single dose P.O. (Grade B, III)

Follow-Up

Patients should be reexamined with serological RPR follow-up at months 1, 2, 3, 6 and 12, then 6 monthly until RPR negative or sero-fast at low titre RPR 1:8 or less

LATE SYPHILIS

Late syphilis is defined as syphilis occurring 2 years after infection and includes late latent syphilis, benign tertiary syphilis (Gumma), cardiovascular syphilis and neurosyphilis

LATE LATENT SYPHILIS

Syphilis of more than 2 years duration; diagnosed by positive serology without any symptoms. All patients should have a thorough clinical examination to exclude cardiovascular and neurological involvement.

Investigations

- CXR
- Lumbar puncture for
 - All those with neurological or ophthalmic signs or symptoms
 - CSF should be sent for biochemical analysis, WBC count and CSF-VDRL test

Patients with neurological symptoms or signs should be referred to a neurologist/physician for a thorough neurological examination to rule out other causes or for further management.

GUMMA (BENIGN TERTIARY SYPHILIS)

Incubation Period: 1 to 45 years with an average of 15 years

Presentation/Findings

Gumma is a destructive granulomatous lesion, commonly presenting with skin or bone lesions. Liver, heart, brain, stomach and the respiratory tract may be affected.

Treponemes can be demonstrated in the tissues by silver impregnation staining or PCR.

CARDIOVASCULAR SYPHILIS

Incubation Period: 15–30 years

Presentation

Although syphilis may affect any large vessel, it is characterized by an aortitis affecting the proximal aorta. The aortitis may cause aortic incompetence (which may be complicated by heart failure), coronary ostial stenosis (presenting as angina), and aortic medial necrosis causing aortic aneurysm.

TREATMENT OF LATE LATENT SYPHILIS, GUMMA OR CARDIOVASCULAR SYPHILIS

Recommended Regimen

- Benzathine penicillin 2.4 MU I.M. weekly for three weeks (three doses) (Grade B, III); OR
 - Procaine penicillin (600,000 units) I.M. daily for 17 days (Grade B, III)
 - Note: If a patient defaults Benzathine penicillin treatment by more than two weeks in between the weekly doses, the whole regime needs to be restarted.
- If a patient defaults Procaine Penicilline injection by more than 1 day, the whole regime needs to be restarted.

Alternative Regimen

For patients with Penicillin allergy,

- Doxycycline 100 mg b.d. P.O. for 28 days (Grade C, IV); OR
- Erythromycin 500mg q.i.d P.O for 28 days (Grade C, IV); OR
- Erythromycin ES 800mg q.i.d P.O for 28 days (Grade C, IV)

Contact Tracing

Examine and investigate sexual partner. Treat if indicated.

Follow-Up

Examine clinically and serologically at 6 monthly intervals for 2 years and thereafter annually till seronegative or stable at a low titre RPR 1:4 or less.

For those in which the non-treponemal test is absent at diagnosis, i.e. RPR: Non-reactive and TPHA positive, clinical follow up is recommended and there is no need in monitoring serology to determine response to treatment

TREATMENT FAILURE IN SYPHILIS

Treated patients should be considered for retreatment if:

- Clinical symptoms or signs persist or recur
- RPR fails to decrease fourfold by one year
- There is a sustained fourfold rise in RPR titre

In the absence of reinfection, CSF examination and re-treatment with 3 weekly doses of IM Benzathine Penicillin 2.4 MU is indicated.

NEUROSYPHILIS

Central nervous system disease may occur during any stage of syphilis. Clinical evidence of neurological involvement warrants CSF examination. Late neurosyphilis includes meningovascular and parenchymatous syphilis.

Diagnosis

- CSF for protein concentration, cell count, CSF VDRL test and *T. pallidum* PCR
- The CSF VDRL test has a low (< 50%) sensitivity but high specificity (with no documented false positives) in the absence of substantial contamination of CSF with blood
- Serum specific treponemal tests should be positive in neurosyphilis

A. MENINGOVASCULAR NEUROSYPHILIS

Incubation Period: 5 to 10 years

Presentation: Headache, vertigo and cranial nerve involvement.

B. PARENCHYMATOUS NEUROSYPHILIS

Incubation Period: 10-20 years

Presentation

Patients may present with general paresis of the insane (brain syndrome) characterized by gradual personality change, ataxia, stroke or ophthalmic symptoms or with tabes dorsalis (spinal cord syndrome) presenting with lightning pain, sensory impairment and mobility problems. Both syndromes are important differential diagnosis in dementia, psychiatric disorders and mobility problems.

Treatment

Recommended Regimen

- Aqueous crystalline penicillin G, 18 - 24 mega units per day, administered 3 – 4 mega units every 4 hourly I.V. for 17-21 days (Grade B, III)

Alternative Regimen

For patients with Penicillin allergy,

- Doxycycline 200 mg b.d. P.O. for 28 days (Grade C. IV); **OR**
- Ceftriaxone 2 gm I.M. (with Lidocaine as diluent) (Grade C, IV); **OR**
- I.V. (with water for injection as diluent NOT Lidocaine) for 10-14 days (*if no anaphylaxis to penicillin*)

Contact Tracing

Examine and investigate sexual partner(s). Treat if indicated.

Follow-Up

Lumbar puncture 6 monthly till CSF cell count is normal

SYPHILIS IN PREGNANCY

Antenatal screening for syphilis with non-treponemal tests (RPR) should be routinely performed on first visit and to repeat at 28 weeks of gestation. Positive results must be confirmed with treponemal tests (TPHA / TPPA / EIA).

Treatment

Penicillin regimen appropriate for the woman's stage of syphilis is recommended.

Doxycycline and tetracycline are contraindicated in pregnancy. Erythromycin should not be used because of the high risk of failure to cure the foetus. If erythromycin is used, paediatricians must be alerted and babies have to be treated prophylactically with penicillin and monitored.

Pregnant women with a history of penicillin allergy should be meticulously interviewed regarding the validity of the history. Skin testing and desensitization can be done if necessary. Women who are treated in the second half of pregnancy are at risk of premature labour and/ or fetal distress if their treatment precipitates a Jarisch-Herxheimer reaction.

They should be advised to seek medical attention if they notice any change in foetal movements or have any contractions following treatment.

Follow-Up

Monthly clinical and serological examination till delivery and thereafter follow-up is as in non-pregnant patients.

CONGENITAL SYPHILIS

Infants should be evaluated if they were born to sero-positive women who:

- Have untreated syphilis
- Were treated for syphilis less than 1 month before delivery
- Were treated for syphilis during pregnancy with a non - penicillin regimen
- Did not have the expected decrease in RPR titre after treatment
- Were treated but had insufficient serologic follow-up during pregnancy to assess disease activity

Evaluation of Infant

- A thorough physical examination
- RPR (compare with mother's titre) / EIA
- FTA-Abs
- CSF analysis for cells, protein and CSF-VDRL test
- Long bones X-ray
- Chest X-ray

Therapy decisions

Infants should be treated if they have:

- Any evidence of active disease
- A reactive CSF-VDRL / FTA-Abs
- An abnormal CSF finding (WBC > 5/ mm³ or protein > 50 mg / dl) regardless of CSF serology
- Serum RPR titre that are at least 4 times higher than their mother's.
- Positive EIA-IgM antibody

Treatment for Congenital Syphilis

- Aqueous Crystalline Penicillin G, 50,000 units/kg/dose 12 hourly for first 7 days then 8 hourly for the following 3-7 days (*Grade B, III*); **OR**
- Procaine Penicillin, 50,000 units/kg daily IM for 10 - 14 days (*Grade B, III*); **OR**
- *IV/IM Ceftriaxone 75 mg/kg (< 30 days old) or 100 mg/kg (>30 days old)

**If more than a day of treatment is missed, the whole course should be restarted*

Infants who should be evaluated but whose follow-up cannot be assured should be treated with a single dose of Benzathine Penicillin, 50,000 units/kg IM.

Follow-Up

Sero-positive untreated infants must be closely monitored at 1, 2, 3, 6, and 12 months of age. RPR should decrease by 3-months of age and usually disappear by 6 months of age.

Treat (with the same regimen as above) if:

- Symptoms and signs persist or recur
- RPR titre increase fourfold or more by 3-months of age
- RPR still positive by 6 months of age
- TPHA still positive by 1 year of age

Treated infants must be monitored clinically and serologically at 1, 3, 6, 12, 18, and 24 months. Lumbar puncture should be repeated 6 monthly till normal.

THERAPY OF OLDER INFANTS AND CHILDREN

After the newborn period, children discovered to have syphilis should have a CSF analysis to rule out congenital syphilis. Any child with congenital syphilis or with neurologic involvement should be treated with Aqueous Crystalline Penicillin, 200,000-300,000 units/kg/day administered as 50,000 units/kg/dose 4-6 hourly for 10 to 14 days (Grade B, III)

SYPHILIS IN HIV - INFECTED PATIENTS

Treatment is the same as for non-HIV infected patients. HIV co-infection with syphilis does not appear to increase the risk of developing a more aggressive course with neurosyphilis, treatment failure or relapse. In early syphilis, HIV infection may be a risk factor for symptomatic early neurosyphilis. HIV test is recommended in all patients with syphilis. HAART also appears to decrease risk of early neurosyphilis in HIV positive patients.

CSF abnormalities (e.g. mononuclear pleocytosis + elevated protein levels) are common in HIV infected patients even those without neurological symptoms.

Several studies have demonstrated that among persons infected with both HIV and syphilis, clinical and CSF abnormalities consistent with neurosyphilis are associated with a CD4 count of <350 cells/mL and/or an RPR titre of >1:32. However unless neurological symptoms are present, CSF examination in this setting has not been associated with improved clinical outcomes.

HIV patients with syphilis should be reevaluated clinically and serologically at 3, 6, 9, 12 and 24 months after therapy to detect any treatment failure.

JARISCH-HERXHEIMER REACTION (JHR)

An acute febrile illness which presents with headache, myalgia, chills and rigors and resolves within 24 hours.

This is common in early syphilis but is usually not important unless there is neurological or ophthalmic involvement or in pregnancy when it may cause fetal distress and premature labour. It is uncommon in late syphilis but can potentially be life threatening if there is involvement of strategic sites (coronary ostia, larynx, nervous system). Prednisolone can reduce the reaction.

Recommendation

In early syphilis	:	Treat with Paracetamol
In Neurosyphilis, Cardiovascular, certain cases of benign tertiary and late latent syphilis	}	Treat with Prednisolone 40-60mg daily for 3 days: Begin 24 hours before treatment and for 2 days after starting treatment.

PROCAINE REACTION

This is due to an inadvertent intravenous injection of procaine penicillin. It is characterized by fear of impending death and may cause hallucinations or fits immediately after injection and lasting less than 20 minutes. Calm and verbal reassurance is required and restraint may be necessary. If fits occur, give Diazepam 10 mg rectally.

- Anaphylactic shock: Facilities for treatment of anaphylaxis should be available as penicillin is amongst the commonest cause.
 - Epinephrine (Adrenaline) 1:1000 I.M. 0.5 ml followed if necessary by I.M./I.V. antihistamine e.g. chlorpheniramine 10mg and I.M./I.V. hydrocortisone 100 mg
- Allergy: Penicillin desensitization may be considered for patients with penicillin allergy.

MANAGEMENT OF PATIENTS WITH HISTORY OF PENICILLIN ALLERGY

Currently, no proven alternative therapies to penicillin are available for treating neurosyphilis, congenital syphilis or syphilis in pregnancy. Therefore, skin testing, with desensitisation, if indicated, should be done for these patients.

Desensitisation should be done in a hospital setting because serious IgE-mediated allergic reactions may occur. The following protocol is recommended. Oral penicillin in increasing concentration is administered every 15 minutes. Sensitisation is completed within 4 hours with a cumulative dose of 1.3 million units of penicillin V. (Refer to Appendix A)

LABORATORY DIAGNOSIS OF SYPHILIS

Demonstration of *Treponema pallidum* from lesions or infected lymph nodes

1. Dark-ground / Dark-field microscopy

- Should be performed by an experienced laboratory technician or doctor
- Success depends on a number of factors, including too little or too much fluid on the slide, improper thickness of the slide or cover slip, etc. Treatment with antibiotics may result in a false negative finding.
- Because of its limited sensitivity, failure to detect *T. pallidum* by this test does not rule out syphilis

2. Polymerase Chain Reaction (PCR)

- Due to limited availability and the time taken to obtain a result, this is not a replacement for dark-field microscopy in the clinic setting.
- In certain circumstances, PCR may be helpful in diagnosis by demonstrating *T. pallidum* in tissue samples, vitreous fluid and CSF.

SEROLOGICAL TESTS

Recommended screening

- RPR has replaced VDRL in both public and private laboratories
- Both non-treponemal RPR and treponemal tests should be routinely performed for all GU medicine clinic attendees
- Antenatal screening with non-treponemal tests RPR should be routinely performed
 - on first visit and at 28 weeks of gestation
 - confirm positive results with treponemal tests (TPHA/TPPA)
- EIA (both IgG and IgM) is ideal screening test for blood bank/transfusion services since it is automated and specific.

- A quantitative RPR should be performed for positive tests to classify/stage the disease, determine the need for and monitoring of treatment
- A RPR titre of >1:16 and/or a positive IgM test indicate active disease and the need for treatment, although serology must be interpreted in the light of the treatment history and clinical findings

Sensitivity and Specificity of laboratory tests for Syphilis

Tests	Sensitivity (%) during stage of infection				Specificity (%)
	Primary	Secondary	Latent	Late	
Nontreponemal tests					
RPR	86	100	98	73	98
Treponemal tests					
TPPA	88	100	100	NA	96
TPHA	86	100	100	99	96
FTA-ABS	84	100	100	96	97
EIA					
IgG	100	100	100	NA	100
IgM	93	85	64	NA	NA
Rapid test					
Immunochromatography/ Latex Agglutination	84.5 - 97.7				92.8 - 98
Dark field microscopy	74 - 86	-	-	-	85 - 100
PCR for <i>T.pallidum</i>	94.7	80	-	-	98.6
CSF VDRL	NA	NA	-	-	100

GONORRHOEA

Aetiology

Neisseria gonorrhoeae

Incubation Period: 2-5 days

Presentation:

Men:

- Urethral discharge (>80%) and dysuria (>50%)
- Anal discharge (12%) or perianal pain (7%)
- Testicular swelling and pain
- Fever, petechial or pustular skin lesions, asymmetrical arthralgia, septic arthritis, tenosynovitis, and very rarely, meningitis or endocarditis, caused by disseminated gonococcal infection (DGI)
- Asymptomatic infection can occur: urethral < 10%, rectum > 85%, pharynx > 90%

Women:

- Altered vaginal discharge, mucopurulent endocervical discharge or contact bleeding, caused by cervicitis
 - Lower abdominal pain and tenderness, caused by pelvic inflammatory disease (PID)
 - Fever, petechial or pustular skin lesions, asymmetrical arthralgia, septic arthritis, tenosynovitis, and very rarely, meningitis or endocarditis, caused by disseminated gonococcal infection (DGI)
 - Asymptomatic infection is common: endocervix > 50%, rectum > 85%, pharynx > 90%
- Transluminal spread of *N. gonorrhoeae* may occur from the urethra to involve the epididymis and prostate in men or from endocervix to involve the endometrium and pelvic organs in women
 - Haematogenous dissemination may also occur from infected mucous membranes, resulting in skin lesions, arthralgia, arthritis and tenosynovitis. Disseminated gonococcal infection is uncommon (<1%).
 - Right hypochondrium pain due to perihepatitis (Fitz-Hugh-Curtis syndrome) is a rare but recognized complication.

Investigations

1. Gram stain of urethral, cervical or rectal exudates
 - Gram negative intracellular diplococci in leucocytes (Gram stain is not appropriate for pharyngeal specimens)
 - Male : Sensitivity 95%, (symptomatic), 50-75% (asymptomatic) Specificity 99%
 - Female : Sensitivity 45 – 65%, Specificity 90%
2. Culture on modified Thayer Martin culture medium performed on all samples
 - Offers a high sensitivity, confirmation of diagnosis, and antibiotic sensitivity testing
 - Sensitivity : 60-70% Specificity 100%
3. Nucleic acid amplification tests (NAATs) and nucleic acid hybridization tests
 - More sensitive than culture If NAAT is used for *N. gonorrhoeae* – a positive test should be confirmed by culture (for antibiotics sensitivity)
 - Can be used as diagnostic / screening tests on non-invasively collected specimens (urine and self-taken vaginal swabs)
 - Caution is required in interpretation of positive results as specificity of NAATs is not 100%
 - Sensitivity : 95 – 96% Specificity 99%
4. Multiplex PCR assays
 - Available for the simultaneous detection of gonorrhoea and chlamydia
 - Sensitivity of 87-98% and specificity of 98 – 100%

Treatment

A. Recommended Regimen

Uncomplicated anogenital infection in adults:

- Ceftriaxone 500* mg I.M. as a single dose (*Grade C, IV*) AND
- Azithromycin 1g PO as a single dose **(*Grade C, IV*)

* dose increased from 250mg following development of resistance in other part of the world

** for synergistic therapeutic effect, reduction of resistance and treatment of *C.trachomatis* co-infection

B. Alternative Regimen

May be used when an infection is known to be sensitive to these antimicrobials or when the regional prevalence of resistance to them is less than 5%:

- Cefixime 400 mg single dose P.O. (*Grade A, Ib*); OR
- Cefotaxime 500 mg I.M. as a single dose (*Grade A, Ib*); OR
- Spectinomycin 2 g I.M. as a single dose (*Grade A, Ib*)

Penicillin, tetracyclines and quinolones are no longer recommended for treatment of gonorrhoea owing to high resistance rates to these antibiotics worldwide. Azithromycin alone is not a recommended treatment for gonorrhoea.

Clinicians using alternative regimens are recommended to regularly review local sensitivity data with their microbiology colleagues.

C. Patients with β -lactam Allergy

Recommended Regimen

- Spectinomycin 2 g I.M. as a single dose (*Grade A, Ib*)

PREGNANCY AND BREASTFEEDING

Pregnant and breastfeeding women should not be treated with quinolone or tetracycline antimicrobials.

Recommended Regimen

- Ceftriaxone 500 mg I.M. and Azithromycin 1 gm P.O as a single dose (*Grade C, IV*); OR
- Spectinomycin 2 gm I.M. as a single dose (*Grade A, I*)

PHARYNGEAL INFECTION

Recommended Regimen

- Ceftriaxone 500 mg I.M. and Azithromycin 1 gm P.O as a single dose (*Grade C, IV*)
- Single dose treatment using ampicillin or spectinomycin has a poor efficacy in eradicating gonococcal infection of the pharynx

GONOCOCCAL CONJUNCTIVITIS

Recommended Regimen

- IM Ceftriaxone 500 mg daily for 3 days; **OR**
- Azithromycin 2 gram P.O single dose, **PLUS**

Doxycycline P.O 100 mg bd for 1 week, **PLUS**
Ciprofloxacin 250 mg OD for 3 days; **OR**

- IM Spectinomycin 2 gm OD for 3 days

DISSEMINATED GONOCOCCAL INFECTION (DGI)

Recommended management is based on expert opinion and accumulated clinical experience

Hospitalize patient and treat with

- Ceftriaxone 1 gm I.M. or I.V. every 24 hours; OR
- Cefotaxime 1 gm I.V. every 8 hours; OR
- Spectinomycin 2 gm I.M. every 12 hours

Therapy should continue for at least 7 days.

CO-INFECTION WITH *CHLAMYDIA TRACHOMATIS*

Up to 35% of men and 40% of women with gonorrhoea have co-infection with *C. trachomatis*.

Screening for *C. trachomatis* should routinely be performed on adults with gonorrhoea or treatment given to eradicate possible co-infection.

Combining effective antimicrobial therapy against *C. trachomatis* with a single dose therapy for gonococcal infection is particularly appropriate when there is doubt that a patient will return for follow-up evaluation.

Advice

Patients should be advised to avoid unprotected sexual intercourse until they and their partner(s) have completed treatment.

Contact Tracing

Male patients with symptomatic urethral infection should notify all partners with whom they had sexual contact within the preceding 3 months

Patients with infection at other sites or asymptomatic infection should notify all partners within the preceding 3 months.

Sexual partners should be treated for gonorrhoea preferably after evaluation for sexually acquired infection.

Follow-Up

Patients should be assessed after treatment:

- To confirm compliance to treatment
- To ensure resolution of symptoms
- To enquire about adverse reactions
- To re-take sexual history to explore the possibility of re-infection
- To pursue partner notification and health promotion

EPIDIDYMO-ORCHITIS**Introduction**

A syndrome consisting of pain, swelling and inflammation of the epididymis and / or the testes.

Etiology

Usually under 35 year of age; most often caused by a sexually transmitted pathogen e.g. *Chlamydia trachomatis* and *N. gonorrhea*.

Over 35 years of age; usually due to non sexually transmitted gram negative enteric organisms e.g. *E.coli*

Anatomical and functional abnormalities are often present in this group and should be further investigated.

Other causes: Mumps, Tuberculosis

Important differential diagnosis

- Torsion of testes – it is a surgical emergency; it should be considered in all patients and be excluded first.
- Trauma

Symptoms

- Present with unilateral pain and swelling of the testes
- Dysuria
- There may be symptoms of urethritis or a urethral discharge or it can be asymptomatic

Signs

- Palpable swelling of the epididymis
- Tenderness on palpation
- There may be erythema / edema of the scrotum, urethral discharge and pyrexia.

Investigations

- Gram stained urethral smear – to look for polymorphs and intracellular gram negative diplococci
- *N. gonorrhoea* culture
- Chlamydia immunofluorescence or NAAT
- Urine FEME and culture

Management

- Scrotal support
- Analgesia – NSAIDs
- Antibiotics :
 - I.M Ceftriaxone 250 mg single dose (Grade B, IIIb), PLUS
 - T. Doxycycline 100 mg bd for 14 days; OR
 - T. Ofloxacin 200 mg bd for 14 days (if enteric organisms suspected)

Complications

1. Hydrocele
2. Abscess formation
3. Infarction of testes
4. Infertility

Partner Notification

- Trace partner/s for screening and treatment
- Abstain from sexual intercourse until patient and partner completed treatment

Follow-Up

If there is no clinical improvement after three days, the diagnosis should be reevaluated. Further follow-up is recommended after two weeks to assess treatment compliance, partner notification and resolution of symptoms.

CHLAMYDIA TRACHOMATIS GENITAL TRACT INFECTION

Chlamydia trachomatis is the commonest bacterial STI and the prevalence is highest in persons aged ≤ 25 years. Asymptomatic infection is common among both men and women.

Aetiology

Chlamydia trachomatis serovars D to K

Incubation Period: 7-21 days

Presentation**In women:**

- Asymptomatic (60-70%)
- Mucopurulent vaginal discharge (30-40%)
- Intermenstrual or post coital bleeding
- Lower abdominal pain
- Acute and chronic symptoms and signs of Pelvic Inflammatory Disease
- Cervicitis

In men:

- Asymptomatic (50-60%)
- Mucopurulent urethral discharge
- Signs of epididymo-orchitis and prostatitis

In both genders:

- Dysuria
- Ano-rectal discomfort + discharge
- Arthralgia
- Pharyngeal infections
- Conjunctivitis

Complications

- Pelvic inflammatory disease (PID)
- Ectopic pregnancy
- Infertility
- Sexually acquired reactive arthritis (SARA)
- Periurethral and perianal abscess

Investigations

Specimens for Testing

In men : urethral swab and/or first void urine
In women : endocervical swab and/or first void urine

Laboratory tests

1. Gram stain

- Increased PMNs (average of ≥ 5 per high power field in urethral smear and ≥ 20 per high power field in endocervical smear, ≥ 10 in first void urine)
- To exclude Gram-negative intracellular diplococci

2. Cell culture

- Considered the gold standard
- Specificity almost 100% but has sensitivity of 40-80%
- Routine use is not recommended

3. Direct Fluorescent Antibody Test (DFAT) for chlamydial antigen

- Sensitivity of 80-90% and specificity of 99%

4. Enzyme Immunoassays (EIA)

- Sensitivity of 20-75%; Specificity 92-100%
- EIA's are being replaced by NAAT's.

5. Nucleic acid amplification tests (NAAT's Grade A, I

- These tests are highly sensitive and specific and are suitable for non-invasive samples such as urine and low vaginal swabs.
- Multiplex PCR assays are available for the simultaneous detection of chlamydia and gonorrhoea.
- Sensitivity of 87-98% and specificity of 98 – 100%
- Consider pharyngeal and rectal swabs depending upon sexual exposure
- The table below shows the relative sensitivities of different laboratory tests for the diagnosis of chlamydial infections in females by specimen type:

	Cervix**	Urine	Vulvovaginal
Culture	50 – 70%	5 – 20%	20 – 30%
DFA	45 – 55%	NA	NA
EIA	40 – 50%	NA	25 – 40%
NAAT	87 – 93%	80 – 90%	80 – 90%

* NA: not applicable

** Endocervical swab is preferable

Treatment

Uncomplicated urethral, endocervical, or rectal chlamydial infections

Recommended (Grade A, I)

- Doxycycline 100 mg b.d. P.O. for 7 days; OR
- Azithromycin 1 g single dose P.O.

Alternative (Grade B, II)

- Ofloxacin 200 mg b.d. or 400 mg daily P.O. for 7 days; OR
- Erythromycin stearate 500 mg q.i.d. P.O. for 7 days; OR
- Erythromycin ethyl succinate 800 mg q.i.d. P.O. for 7 days

Regimen for Pregnant Women (Grade A, I)

- Erythromycin stearate 500 mg q.i.d. P.O. for 7 days
- Erythromycin ethyl succinate 800 mg q.i.d. P.O. for 7 days
- Amoxicillin 500 mg tds P.O. for 7 days
- Azithromycin 1 gm P.O stat

Chlamydial Infections among Infants

- Prenatal screening of pregnant women may prevent chlamydial infection among neonates.
- Pregnant women aged <25 years are at high risk for infection.
- *C. trachomatis* infection of neonates results from perinatal exposure to the mother's infected cervix.
- Initial *C. trachomatis* perinatal infection involves the mucous membranes of the eye, oropharynx, urogenital tract, and rectum and may be asymptomatic
- *C. trachomatis* infection in neonates is most frequently manifested as conjunctivitis that develops 5-12 days after birth.
- *C. trachomatis* may cause a subacute, afebrile pneumonia with onset at ages 1-3 months

Contact Tracing

- Sexual partners of patients with chlamydial infection must be tested and if positive must be treated likewise.
- If testing is not available, or if the partners are unwilling for examination, they must be treated empirically
- Neonates of pregnant mothers with *Chlamydial* infections must be screened

Advice

Patients should abstain from sexual intercourse for 7 days until after they and their sexual partners have completed treatment.

NON-GONOCOCCAL URETHRITIS (NGU)**Etiology**

Urethritis is characterized by urethral inflammation, which results from infectious and non-infectious conditions and may or may not present with symptoms. Urethritis is described as either gonococcal or nongonococcal (NGU). Mucopurulent cervicitis is the female equivalent of male NGU, with approximately 40% of cases being due to infection with *Chlamydia trachomatis*

Common NGU organisms include:

- *Chlamydia trachomatis*,
- *Mycoplasma genitalium*
- *Ureaplasma urealyticum*,
- *Trichomonas vaginalis*

Presentation

- Urethral discharge, which may be muco-purulent or mucoid and may only be present on urethral milking
- Dysuria
- Urethral itching
- Penile irritation
- Urinary frequency
- Asymptomatic

Other clinical signs include balanitis, balanoposthitis, epididymo-orchitis and SARA (Sexually Acquired Reactive Arthritis)

Investigations

All patients who have confirmed or suspected urethritis should be tested for *N.gonorrhoeae* and *C.trachomatis*. Testing for Chlamydia is strongly recommended with high sensitivity and specificity test methods (e.g. NAATs).

1. Gram stain on urethral smear

- Gram stained urethral smear containing 5 or more PMNL per high power field in urethral smear
- There is little justification in performing urethral microscopy in asymptomatic men (IIb)

2. Chlamydia Antigen Immunofluorescence (IF) test

- Urethral swab using Dacron stick

3. Urethral swab for gonococcal culture or NAAT

- All patients should have a test for *N.gonorrhoeae* either by culture or NAAT (B, IIa)

4. Urine NAAT for *Chlamydia* and *N. gonorrhoeae*

- If NAAT is used for *N. gonorrhoeae* a positive test should be confirmed by culture
- *C. trachomatis* should also be sought. It should be noted that even a NAAT will miss between 3% and 10% of infections (B, III)

5. Urethral swab microscopy (wet mount) for *T. vaginalis*

Persons who have been diagnosed with a new STIs should receive testing for other infections, including syphilis and HIV

Advice

The following should be discussed and clear written information provided: (Grade C, IV)

- An explanation of the causes of urethritis, including non-infective causes, and possible short term and long term implications
- The side-effects of the treatment and the importance of complying fully with it
- The importance of their sexual partner(s) being evaluated and treated
- Abstain from sexual intercourse until he has completed therapy and his partner(s) have been treated.
- Advice on safer sex and consistent condom use
- Follow-up is recommended

Treatment**Recommended Regimens (Grade A)**

- Azithromycin 1g orally in a single dose (Ib); OR
- Doxycycline 100mg b.d orally for 7 days (Ib)

Alternative Regimens (Grade A)

- Erythromycin stearate 500mg b.d for 14 days (Ib); OR
- Ofloxacin 200mg b.d or 400mg once a day for 14 days (Ib); OR
- Erythromycin ethylsuccinate 800mg orally q.i.d for 7 days; OR
- Levofloxacin 500mg orally once daily for 7 days

Contact Tracing

All sexual partners within the preceding 6 months should be referred for evaluation, testing, and empiric treatment with a drug regimen effective against Chlamydia.

Follow-Up for Patients with NGU

- Follow-up after 2-3 weeks
- Patients who remain symptomatic, who have not completed their medication or who have had unprotected sexual intercourse with an untreated partner should be asked to return to the clinic and re-treated with appropriate contact tracing (C, IV)
- A test of cure in NGU in an otherwise asymptomatic individual is not recommended (B, III)

Persistent Urethritis in Male Patients

Persons who have persistent or recurrent urethritis can be retreated with the initial regimen if they did not comply with the treatment regimen or if they were re-exposed to an untreated sexual partner.

For symptomatic patients or an observable discharge

- Azithromycin 500mg STAT then 250mg o.d for the next 4 days (Grade B, IIa), PLUS
- Metronidazole 400 mg b.d for 5 days (Grade C, IV); OR
- Erythromycin 500mg q.i.d for 3 weeks (Grade A, Ib), PLUS
- Metronidazole 400mg b.d for 5 days (Grade C, IV)

Cervicitis

Cervicitis is frequently asymptomatic, but some women complain of an abnormal vaginal discharge and intermenstrual or postcoital bleeding.

Two major diagnostic signs include:-

1. A purulent or mucopurulent endocervical exudate visible in the endocervical canal (commonly referred to as mucopurulent cervicitis or cervicitis)
2. Sustained endocervical bleeding easily induced by gentle passage of a cotton swab through the cervical os.

Investigations

Because cervicitis may be a sign of upper genital tract infection (endometritis), women who seek medical treatment for a new episode of cervicitis should be assessed for signs of PID and should be tested for *C. trachomatis* and for *N. gonorrhoea*. Women with cervicitis should also be evaluated for the presence of BV and trichomoniasis, and if these infections are detected, they should be treated.

Treatment

Several factors should affect the decision to provide presumptive therapy for cervicitis or to await the results of diagnostic tests. Treatment with antibiotics for *C. trachomatis* should be provided for those women at increased risk for this common STI especially if follow-up cannot be ensured and if a relatively insensitive diagnostic test is used in place of NAAT. Presumptive therapy should be considered for the following :-

- Aged 25 or under
- Those with new or multiple sexual partners
- Those who engage in unprotected sex

Recommended regimens for presumptive treatment

- Azithromycin 1g orally in a single dose; OR
- Doxycycline 100mg P.O b.d for 7 days

Special considerations

HIV infection

- Patients who have cervicitis and are also infected with HIV should receive the same treatment regimen as those who are HIV negative
- Treatment of cervicitis in HIV-infected women is vital because cervicitis increases HIV shedding
- Treatment of cervicitis in HIV-infected women reduces HIV shedding from the cervix and might reduce HIV transmission to susceptible sexual partners

PELVIC INFLAMMATORY DISEASE

Aetiology

PID is usually the result of infection ascending from the endocervix causing endometritis, salpingitis, parametritis, oophoritis, tuboovarian abscess and/or pelvic peritonitis.

The aetiology of PID is often polymicrobial. Common organisms include :-

- a) Sexually transmitted organisms especially *Neisseria gonorrhoea* and *Chlamydia trachomatis*
- b) Non sexually transmitted pathogen (e.g. anaerobes, *Gardnerella vaginalis*, *Haemophilus influenzae*, enteric Gram-negative rods, *Streptococcus agalactiae*, Cytomegalovirus (CMV), *Mycoplasma hominis* and *Ureaplasma urealyticum*

Presentation:

- Lower abdominal pain which is typically bilateral
- Deep dyspareunia
- Abnormal vaginal bleeding, including post coital, intermenstrual and menorrhagia
- Abnormal vaginal or cervical discharge which is often purulent
- Adnexal tenderness on bimanual vaginal examination
- Cervical motion tenderness on bimanual vaginal examination
- Fever (>38°C)

Diagnosis:

- PID is usually symptomatic, but may be asymptomatic
- Endocervical swabs for microscopy, culture and sensitivity (MCS) for gonorrhoea and chlamydia in the lower genital tract are highly recommended.
- Other causes of lower abdominal /pelvic pain such as acute appendicitis and urinary tract infection have to be excluded.

Further Investigations

All sexually active patients should be offered:

- a pregnancy test
- screening for sexually transmitted infections including HIV

Consider ultrasound to rule out other pathology including tuboovarian abscess

Management

Empirical treatment should be started immediately if there are any suspicions of PID because of the lack of definitive clinical diagnostic criteria and because the potential consequences of not treating of PID are significant.

1. Outpatient treatment is effective for patients with mild to moderate PID.

2. Hospital admission (preferably under gynaecological care) for parenteral therapy, observation, further investigation and / or possible surgical intervention should be considered in the following situations:
 - I. a surgical emergency cannot be excluded
 - II. lack of response or intolerance to oral therapy
 - III. clinically severe disease (pyrexia > 38°C, signs of peritonitis)
 - IV. presence of a tuboovarian abscess
 - V. pregnancy
3. Appropriate analgesia should be provided.

Treatment

The following antibiotic regimens are evidence based and of similar efficacy.

Outpatient Regimens:

- Oral ofloxacin 400mg twice daily PLUS oral metronidazole 400mg twice daily for 14 days (Grade A, Ib)
- I.M. ceftriaxone 500 mg single dose followed by oral doxycycline 100mg twice daily PLUS metronidazole 400mg twice daily for 14 days (Grade A, Ib)
- I.M. ceftriaxone 500 mg single dose followed by azithromycin 1 g/week for 2 weeks (Grade A, Ib)

Do not replace intramuscular ceftriaxone with an oral cephalosporin (e.g. cefixime) because there is no clinical trial evidence to support its use, and tissue levels are likely to be lower which might impact on efficacy.

Inpatient Regimens:

Intravenous therapy should be continued for 24 hours after clinical improvement and then switched to oral.

- I.V. ceftriaxone 2g daily PLUS I.V. doxycycline 100mg b.d (oral doxycycline may be used if tolerated) followed by oral doxycycline 100mg b.d PLUS oral metronidazole 400mg b.d for a total of 14 days (Grade A, Ib)
- I.V. clindamycin 900mg t.d.s PLUS I.V. gentamicin (2mg/kg loading dose) followed by 1.5mg/kg t.d.s [a single daily dose of 7mg/kg may be substituted] followed by either oral clindamycin 450mg q.i.d OR oral doxycycline 100mg b.d PLUS oral metronidazole 400mg b.d to complete 14 days (Grade A, Ib)
- I.V. ciprofloxacin 200mg b.d PLUS I.V. (or oral) doxycycline 100mg b.d PLUS I.V. metronidazole 500mg t.d.s for 14 days (Grade B, III)

Pregnancy and Breastfeeding

PID in pregnancy is associated with an increase in both maternal and fetal morbidity, therefore parenteral therapy is advised although none of the suggested evidence based regimens are of proven safety in this situation.

Recommended regime:

- I.M. ceftriaxone PLUS oral / I.V. erythromycin, with the possible addition of oral / I.V. metronidazole 500mg t.d.s in clinically severe disease (Grade C, IV)

Surgical Management

Laparoscopy may help early resolution of the disease by dividing adhesions and draining pelvic abscesses but ultrasound guided aspiration of pelvic fluid collections is less invasive and may be equally effective.

General advice

- Patients should be advised to avoid unprotected intercourse until they, and their partner(s), have completed treatment and follow-up (Grade C, IV)
- A detailed explanation of their condition with particular emphasis on the long term implications for the health of themselves and their partner(s) should be provided and reinforced with clear and accurate written information (Grade C, IV)
- When giving information to patients, the clinician should consider the following:
 1. An explanation of what treatment is being given and its possible adverse effects
 2. That following treatment fertility is usually maintained but there remains a risk of future infertility, chronic pelvic pain or ectopic pregnancy
 3. Clinically more severe disease is associated with a greater risk of sequelae
 4. Repeat episodes of PID are associated with an exponential increase in the risk of infertility
 5. Future use of barrier contraception will significantly reduce the risk of PID

N.B The decision to remove an intrauterine device (IUD) in a woman with suspected PID needs to be balanced against the risk of pregnancy in those who have had otherwise unprotected intercourse in the preceding 7 days. Hormonal emergency contraception may be appropriate for some women in this situation.

Sexual Partners

- Current male partners of women with PID should be contacted and offered health advice and screening for gonorrhoea and chlamydia. (Grade C, IV)
- Gonorrhoea or chlamydia diagnosed in the male partner should be treated appropriately and concurrently with the index patient. (Grade C, IV)
- Broad spectrum empirical therapy should be offered to male partners e.g. I.M. ceftriaxone 500 mg single dose with azithromycin 1g single dose (Grade C, IV)
- Partners should be advised to avoid intercourse until they and the index patient have completed the treatment course (Grade C, IV)

Follow Up

Review at 72 hours is recommended. (Grade C, IV). Lack of clinical improvement suggests the need for further investigation, parenteral therapy and / or surgical intervention.

Further review in 2 – 4 weeks (Grade C, IV) after therapy to ensure:

- Adequate clinical response to treatment
- Compliance with oral antibiotics
- Screening and treatment of sexual contacts
- Awareness of the significance of PID and its sequelae
- Repeat pregnancy test, if clinically indicated

OPHTHALMIA NEONATORUM**Definition**

Conjunctivitis which occurs in the first 4 weeks of life in a neonate

Aetiology

- i) *Chlamydia trachomatis* (the commonest causative organism)
- ii) *Neisseria gonorrhoeae*
- iii) Other bacterial agents : *Staphylococcus aureus*, *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Pseudomonas* and Enteric gram negative organism
- iv) Viral (HSV 1 & 2)

Incubation Period

C. trachomatis conjunctivitis presents at 5-14 days of life

Gonococcal conjunctivitis presents at 2-5 days of life (range 0-28 days)

Presentation / Findings

An initial watery conjunctival exudate may rapidly become purulent, thick and blood stained.

The conjunctiva and eyelid are oedematous. If untreated, there may be development of keratitis, iridocyclitis, corneal ulceration, perforation, pannus formation, scarring, and eventually blindness.

In some infants with Chlamydia-associated conjunctivitis, the infection persists for a longer period and the babies later develop pharyngitis and pneumonia if left untreated.

Investigations

- i) Gram stain smear of conjunctival discharge or scraping
- ii) Culture for *N. gonorrhoeae* – Sensitivity of 60 - 70% and specificity of 100%
- iii) Direct Immunofluorescent test for *C. trachomatis* – Sensitivity of 50 - 90% and specificity of 95 - 99%
- iv) Perform FTA Absof the infant to exclude associated congenital syphilis
- v) PCR for *N. gonorrhoeae* & *C. trachomatis* – Sensitivity of 87-98% and specificity of 98-100%

Treatment

Gonococcal Ophthalmia

Systemic

- Ceftriaxone 50mg/kg I.V. or I.M. daily for 3-7 days (Grade C, IV); OR
- Cefotaxime 50mg/kg/day I.V. or I.M. in two divided doses for 3-7 days (Grade C, IV)

Disseminated infection – treat for 7 days

Meningitis – treat for 10-14 days

Local

- Irrigate eyes with sterile normal saline at least hourly to eliminate discharge
- Topical antibiotics are optional

Chlamydial Ophthalmia

Systemic

1. First week of life

- Erythromycin 20 mg/kg/day P.O. in four divided doses for 14 days (< 2 kg)
- Erythromycin 30 mg/kg/day P.O. in four divided doses for 14 days (> 2 kg); OR
- Erythromycin ES 50mg/kg/day P.O. in four divided dose for 14 days

2. More than 1 week of life

- Erythromycin 40 mg/kg/day P.O. in four divided doses for 14 days

Local

- Tetracycline 1% ointment 6 hourly for 7-14 days

Systemic treatment is essential. Refer to ophthalmologist to assess for ocular complications.

Advice

Careful handling of neonate to prevent direct spread to others

Contact Tracing

Screen both parents for STIs. The mother should be treated on epidemiological grounds.

Follow-Up

On discharge, infants should be seen at 2 weeks to have a repeat eye swab gram stain and culture. If clinical symptoms and signs persist, to consider a second course of treatment.

GENITAL HERPES

Aetiology

Herpes simplex virus type I or type II (HSV-1, the usual cause of oro-labial herpes)

Incubation Period: 2 – 21 days

Presentation

Symptoms:

- Painful vesicles, ulceration, dysuria, vaginal or urethral discharge
- Systemic symptoms e.g. fever and myalgia
- Patient may be asymptomatic, and the disease unrecognized
- Rarely, systemic symptoms may be the only evidence of infection
- Local and systemic symptoms are commoner and more severe in primary than in initial or recurrent disease

Findings:

- Blistering and ulceration of the external genitalia ($\pm$ cervix/rectum)
 - First episode – primary / non–primary
 - Recurrent
 - Atypical
- Inguinal lymphadenopathy

Complications:

- Autonomic neuropathy, resulting in urinary retention
- Aseptic meningitis

FIRST EPISODE PRIMARY

- First noted symptoms with no previous exposure to HSV and sero-negative for HSV 1 & HSV-2
- Lesions usually heal within a month
- Following primary infection, the virus becomes latent in local sensory ganglia, periodically reactivating to cause symptomatic lesions or asymptomatic, but infectious, viral shedding
- Median recurrence rate after a symptomatic first episode is 0.34 recurrences / month for HSV-2 and 0.08 recurrences / month for HSV-1
- Recurrence rates decline over time in most individuals, although this pattern is variable
-

	First episode primary
Symptoms & signs	– Vesicular-ulcerative lesions – Lesions multiple bilateral, different stages – Tender lymphadenopathy (90% & bilateral)
Complications	– Occur in 40-70% – Difficulty urinating (10-15%) – Systemic illness (25%) e.g. headache, photophobia, neck stiffness – Rarely disseminate

FIRST EPISODE NON-PRIMARY

- First noted symptoms, previous exposure to HSV, sero-positive to other HSV type
- Symptoms and signs less severe as compared to first episode primary
- Lesions usually heal within 1-2 weeks

RECURRENT HERPES GENITALIS

- Recurrence of symptoms and signs
- Milder than primary infection
- Certain factors may precipitate an attack such as trauma, menstruation, concurrent infection or fever, immune suppression, stress, and UV light
- Lesions usually heal within 1-2 weeks

	Symptomatic recurrent episode
Symptoms & signs	– Prodromal symptoms (itching, tingling, neuralgia) may occur hours or days before episode – 70% – Lesions localised and unilateral – Tender lymphadenopathy (25% & unilateral)
Complications	– Systemic symptoms uncommon

ATYPICAL HERPES GENITALIS

- Atypical recurrent genital herpes is frequently encountered
- Misdiagnosis is common as a consequence
- Present as excoriation marks, folliculitis, fissure, or erythema over the genital region

ASYMPTOMATIC HERPES

- 80% of patients have no symptoms following infection
- Shedding of virus may occur from intact epithelial surface in the absence of symptoms
- Virus is shed intermittently 5 – 20% days per year (17 to 72 days per year by PCR testing)
- Responsible for transmission of HSV in majority of cases (70%)
- Majority of individuals with asymptomatic HSV-2 infection subsequently develop symptomatic lesions

Investigations

Laboratory tests (Essential as it influences management, prognosis and counseling)

- Virus antigens by DFA; Sensitivity 49-80%, Specificity 100%
- HSV nucleic acid (DNA) by PCR
- Viral culture (culture live virus from lesion) – sensitivity 75% (primary), 50% (recurrent); specificity (100%). Culture sensitivity is highest if taken from vesicular lesion, primary lesion and base of ulcer
- Tzanck test – Sensitivity of 40 - 60%
- Biopsy
- Serology (Group specific antibodies / type specific antibody)

1. Virus antigens

- Direct examination of fluid from (early & atypical) blister for HSV Ag using IFAT
- Differentiate HSV-1 from HSV-2
- Sensitivity 49 – 80% (Higher sensitivity with early lesions)
- Specificity High (if type specific reagents used)
- False negative may occur e.g. due to inadequate smear collection, late lesions or due to anti-viral therapy

2. HSV nucleic acid by polymerase chain reaction (Grade A, Ib)

- Useful for all stages but not widely available
- Detects HSV DNA, type specific

- Increases HSV detection rates in muco-cutaneous swabs by 11 – 71% compared with viral culture
- Sensitivity and specificity >90%
- Test of choice for the diagnosis of HSV infection of the CNS / deep seated infections e.g. lung

3. Viral culture

- Gold standard for diagnosing HSV infection (early & atypical)
- Fluid & cells from base of lesion with pre-moistened swab in viral transport medium
- Sensitivity: Highest in vesicular stage
70 – 80% in first episode
50% in ulcers of recurrent lesions
25% in crusted lesions
- Specificity virtually 100% (if typing performed)
- Negative culture does not exclude infection

Rate of culture positivity for HSV according to stage of lesions

Stage of Lesion	Percentage
Vesiculopapular (early)	82%
Ulcer (mid stage)	43%
Crusted (late)	20%

Ref: Herpes Simplex Virus Detection from Genital Lesions: a Comparative Study Using Antigen Detection (HerpChek) And Culture, Richard W. Cone et. al, J of Cl Microbiology, July 1993

4. Tzanck test / smear

- Easy and rapid bedside test in the presence of classical clinical features
- However, it is subjective and not a definitive test
- Specimen obtained from base of ulcer, not vesicle fluid
- Air dry, fix in 95% ethanol & apply Giemsa stain
- Look for multinucleated giant cells
- Sensitivity 40 – 60%
- Does not differentiate HSV from VZV
- Insensitive & nonspecific & should not be relied on for diagnosis of HSV infection

5. Biopsy

- For persistent ulcers and verrucous lesions in immunosuppressed patients
- Typical histological changes but does not differentiate HSV from VZV

6. Serology

- Antibodies of IgG class persist indefinitely but of limited significance in diagnosis.

7. Type specific serology tests (TSSTs)

- Type-specific immune response can take 8-12 weeks to develop following primary infection
- Uses type-specific EIAs based on glycoprotein G (gG1, gG2) or western blot assays
- Detects HSV-specific glycoprotein G2 for HSV2; G1 for HSV1
- Has a sensitivity of 80-98% and a specificity of 96% or more
- Caution is needed in interpreting results because even highly sensitive and specific assays have poor predictive values for **low prevalence** populations

Serology test is not routinely recommended in asymptomatic patients but is indicated in: (IUSTI European Guidelines 2010)

- History of recurrent or atypical genital herpes when direct virus detection is negative*
- First episode genital herpes where differentiating between primary and established infection guides counseling and management*
- Sexual partners of patients with genital herpes where concerns are raised about transmission*

Comparison of detection methods for HSV in clinical lesions				
	Sensitivity	Specificity	Viral typing	Comments
Tzanck smear	Low	Low	No	- Shows giant cells: presumptive evidence of infection - Unreliable in patients with incompetent CMI
Viral culture	High	High	Yes	- Ideal test - Sensitivity declines as lesion heals
Antigen detection (IFAT / EIA)	High	High	Yes (IFAT)	Rapid but sensitivity declines as lesion heals
PCR	Highest	High	Yes	- Rapid but expensive - Useful even in late lesions - Test of choice in examination of CSF body aspirates

Treatment

A) First Episode of Genital Herpes

Supportive therapy

- Saline/diluted potassium permanganate Sitz bath/dabs
- Analgesia (systemic or local e.g. lidocaine gel)
- Treat any secondary infection

Specific antiviral therapy

- Oral antiviral drugs indicated within 5 days of the start of the episode and while new lesions are still forming
- Acyclovir, valaciclovir and famciclovir all reduce severity & duration of clinical attacks (Grade A, Ib)
- Antiviral therapy does not alter the natural history of the disease
- None of the antivirals eradicate infection or latent virus
- Topical antivirals are less effective than oral agents and not recommended (Grade C, IV) due to the association with acyclovir resistant strain
- Intravenous therapy is indicated when patient cannot swallow or tolerate oral medication because of vomiting
- Addition of topical antivirals to oral treatment is of no benefit

Recommended regimens for first episode of genital herpes - all for 5 days

- Acyclovir 200 mg 5 times/day or 400 mg t.d.s P.O. (Grade A, Ib)
- Valaciclovir 500 mg to 1g b.d. P.O. (Grade A, Ib)
- Famciclovir 250 mg t.d.s. P.O. (Grade A, Ib)

BASHH guidelines recommend > 5 days of treatment if:

- New lesions continue to form
- Symptoms & signs are severe
- Patient also has HIV

Management of complications

- Hospitalization may be required for urinary retention, meningism, and severe constitutional symptoms.
- If catheterization is required, suprapubic route is preferred
 - To prevent theoretical risk of ascending infection
 - To reduce the pain associated with the procedure

- To allow normal micturition to be restored without multiple removals and recatheterizations

B) Recurrent Genital Herpes

- Recurrences are self-limiting and generally cause minor symptoms
- Management decisions should be made in partnership with the patient according to the severity of symptoms and the frequency of recurrence
- Strategies include
 - Supportive therapy only
 - Episodic antiviral treatment
 - Suppressive antiviral therapy

Supportive therapy:

- Saline/diluted potassium permanganate Sitz bath/dabs
- Analgesia (systemic or local e.g. lidocaine gel)
- Treat any secondary infection

Episodic antiviral therapy:

- If severe and/or prominent prodromal symptoms
- Patient provided with a supply of antiviral drug and to initiate during prodrome or early in attack (within 1 day)
- Reduces severity & duration by a median of 1-2 days (Grade A, Ib)

Recommended regimens for episodic therapy (Grade A, Ia):

- Acyclovir
 - 400 mg t.d.s. P.O. for 5 days; OR
 - 200 mg 5 times / day P.O. for 5 days; OR
 - 800 mg b.d. P.O. for 5 days; OR
 - 800 mg t.d.s. P.O. for 2 days (short course)
- Valaciclovir
 - 500 mg b.d. P.O. for 5 days; OR
 - 1 g daily P.O. for 5 days; OR
 - 500 mg b.d. P.O. for 3 days (short course)
- Famciclovir
 - 125 mg b.d. P.O. for 5 days; OR
 - 1 g b.d. P.O. for 1 day (short course)

Head to head studies found no advantage of any one particular antiviral or duration of treatment

C) Suppressive Antiviral Therapy

- If very frequent (≥ 6 per year), severe, prolonged, or with psychosocial problems

- Choice of treatment depends on patient compliance and cost
- Reduce (by 70-80%) the number of recurrences with improvement of QOL
- Does not eliminate subclinical viral shedding
- Discontinue after 12 months to assess ongoing frequency of recurrences because episodes diminish with time. The minimum period of assessment should include two recurrences. Patients who continue to have unacceptably high rates of recurrence may restart therapy
- Safety and resistance data on patients on long-term intermittent therapy with acyclovir now extend to over 18 years of continuous surveillance and 6 years for long term continuous therapy
- Daily suppressive treatment with valaciclovir 500mg daily decreases the rate of HSV-2 transmission in discordant, heterosexual couples in which the source partner has a history of genital HSV-2 infection. Such couple should be encouraged to consider suppressive antiviral therapy as part of a strategy to prevent transmission, in addition to consistent condom use and avoidance of sexual activity during recurrences (Corey L et al. N Eng J Med 2004;350(1):11-20) (Grade A, Ib)

Recommended regimens for suppressive therapy (Grade A, Ib):

- Acyclovir
 - 400 mg b.d. P.O.; OR
 - 200 mg q.i.d P.O.
- Valaciclovir
 - 500 mg once daily P.O. ; OR
 - 1 g once daily P.O.
- Famciclovir
 - 250 mg b.d. P.O. ; OR
 - 500 mg daily P.O.

If breakthrough recurrence occurs on standard suppressive dosage, then increase dosage as per episodic treatment

Severe disease / complication that needs hospitalization
(disseminated infection, pneumonitis, hepatitis, or CNS complication e.g. meningitis or encephalitis)

IV acyclovir 5-10mg/kg 8 hourly for 2-7 days or until clinical improvement followed by oral to complete at least 10 days total treatment.

D) Asymptomatic Viral Shedding

- Occurs in individuals with genital HSV-1 and HSV-2
- Occurs most commonly in
 - Patients with genital HSV-2 infection
 - In the first year after infection
 - In individuals with frequent symptomatic recurrences
- Is an important cause of transmission
- May be reduced by acyclovir 400 mg twice daily, valacyclovir 500 mg or 1 gm daily, famciclovir 250 mg bd (acyclovir = valacyclovir > famciclovir) (Grade A, 1b)

Counseling

- To both the infected person and partner
- Help cope with infection and prevent sexual and perinatal transmission
- Should cover
 - Natural history of genital herpes
 - Use of antiviral drugs for symptom control
 - Risks of transmission by sexual and perinatal means
 - Abstain from sexual contact during lesional recurrences or prodromes
 - Transmission may occur as a result of asymptomatic viral shedding
 - Seropositive patients with unrecognized recurrences can be taught to recognize symptomatic episodes & prevent onward transmission
 - Use of condoms is encouraged although the efficacy to prevent sexual transmission has not been formally assessed
 - Pregnancy issues for both men and women

MANAGEMENT OF GENITAL HERPES IN PREGNANCY

- Categorized into management of first episodes and recurrent episodes
- Accurate clinical classification is difficult. Viral isolation, typing & testing of paired sera may be helpful
- Essential to prevent/reduce neonatal herpes

I) First Episode Genital Herpes

First and second trimester acquisition

- Acyclovir is not licensed for use in pregnancy; however, there is substantial clinical experience supporting its safety i.e. the benefits of antiviral therapy outweigh the risk of withholding treatment (Pregnancy category B)

- Management is in line with clinical condition using either oral or intravenous acyclovir in standard doses
- Vaginal delivery should be anticipated (Grade C, IV)
- Continuous acyclovir in the last 4 weeks of pregnancy reduces risk of both clinical recurrence at term and delivery by caesarean section (CS) (Grade B, Ib)

Third trimester acquisition

- If a true first episode is confirmed, CS should be considered for all women, particularly those developing symptoms after 34 weeks of gestation, as the risk of viral shedding is very high
- CS for the prevention of neonatal herpes has not been evaluated in randomized controlled trials and may not be completely protective against neonatal herpes
- If vaginal delivery is unavoidable, acyclovir treatment of mother and baby may be indicated; prolonged rupture of membrane and invasive procedure e.g. scalp electrodes should be avoided

II) Recurrent Genital Herpes

- Sequential cultures during late pregnancy do not predict viral shedding at term
- If there are no genital lesions at delivery, CS to prevent neonatal herpes should not be performed
- Symptomatic recurrences during the third trimester are likely to be brief; vaginal delivery is appropriate if no lesions are present at delivery
- Continuous acyclovir 400 mg t.d.s. P.O. in the last 4 weeks of pregnancy may reduce the risk of clinical recurrence at term and the need for CS (A, Ia). Acyclovir reduces, but does not eliminate, viral shedding
- Continuous acyclovir in the last 4 weeks of pregnancy may be cost-effective compared with no therapy or with CS.

III) Genital Lesions at the Onset of Labour

- Current practice is for delivery by CS, despite lack of evidence for its effectiveness
- The risks of vaginal delivery for the fetus are small and must be set against risks to the mother of CS

Prevention of Acquisition of Infection

- Maternal risk of HSV acquisition in pregnancy varies geographically and local epidemiological surveillance should guide strategy for prevention
- All women should be asked at their first antenatal visit if their partner/s have ever had GH
- Asymptomatic female partners of men with GH should be strongly advised not to have sex during recurrences. Conscientious use of condoms throughout pregnancy, especially the third trimester, may reduce the risk of acquisition, but this is unproven
- Pregnant women should be advised of the risk of acquiring HSV-1 as a result of oro-genital contact
- Identifying susceptible women by means of type-specific antibody testing has not been shown to be cost-effective
- All women, not just those with a history of GH, should undergo careful vulval inspection at the onset of labour to look for clinical signs of herpes infection
- Mothers, staff, and other relatives/friends with active oral lesions should be advised about the risk of postnatal transmission

Genital Herpes and Perinatal Infection

Infants exposed	At birth	– Follow-up – Some culture mucosa but if asymptomatic, the risk is low
	Near term	– Risk of neonatal herpes – Treat with acyclovir
Neonatal herpes	IV ACV 20 mg/kg 8hrly	– 21 days for disseminated / CNS disease – 14 days for disease limited to skin & mucous membranes

GENITAL HERPES IN HIV PATIENTS

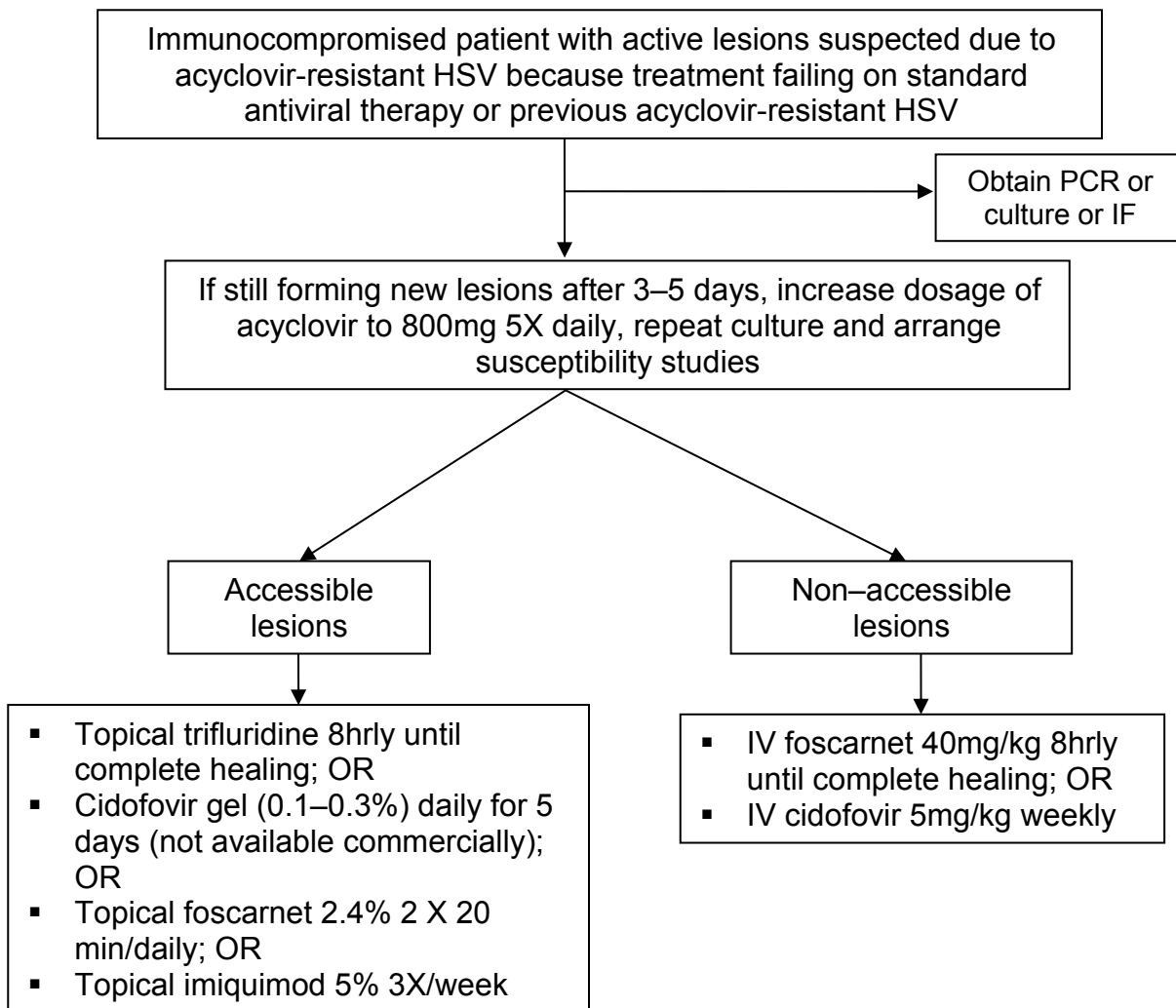
- HSV infection unusually severe / frequently recurrent / prolonged / unusual site / multi-focal
- Genital herpes caused by HSV-2 doubles risk of HIV infection through sexual transmission

Treatment guidelines for HIV infected patients

Episodic	Suppressive
Recommended Regimen	
Episodic (Grade A, Ib) (Grade C, IV) Acyclovir 400mg t.d.s for 5 – 10 days	Acyclovir 400 – 800mg 2–3X per day
Alternative Regimen	
- Acyclovir 200mg 5X per day for 5 – 10 days - Famciclovir 500mg b.d. 5 – 10 days - Valaciclovir 1g b.d. 5 – 10 days	- Famciclovir 500mg b.d. - Valaciclovir 500mg b.d.

Treatment Guidelines Acyclovir-Resistant HSV (lesions persist) in HIV

Recommended Regimen
- Foscarnet 40mg/kg IV 8hrly until healed - Cidofovir 5mg/kg IV once a week until healed
Alternative Regimen
- Topical imiquimod 5% 3 times/week - Topical trifluridine tds. - Topical cidofovir gel for 5 days

Flowchart: Management of acyclovir-resistant HSV in HIV

Adopted from IUSTI European Guidelines 2010

Advice

- Detailed explanation of condition with long term implications for health of themselves and partner
- Reinforced with clear and accurate written information
- Use of condoms may prevent transmission of HSV to uninfected partners and should be encouraged

Contact Tracing

- Effective way of detecting individuals with unrecognized disease
- May clarify whether a partner is infected or not (using type-specific antibody testing if necessary). This may help reduce anxiety about transmission or reinforce need to reduce the risk of transmission

- Awareness of diagnosis in a partner or ex-partner may prevent further onward transmission

Follow-Up

- Weekly until ulcers are healed

GENITAL WARTS

Aetiology

- Benign squamous epithelial tumour caused by *Human papilloma virus* (HPV)
- Genotypes 6 & 11 causes > 90% of cases
- Genotypes 16 & 18 – usually leads to subclinical lesions associated with intraepithelial neoplasia (IN) and anogenital cancer
- HPV types infecting anogenital region can be divided into 2 groups :

High risk HPV (Oncogenic)

16,18,26,31,33,35
39,45,51,52,53,56
58,59,66,73,82

Low risk HPV

6,11,40,42,43,44
54,61,72 & 81

Incubation Period: 2 weeks – 8 months (usually 3 months)

Presentation:

- Very infectious – more than 75% of sexual partners develop warts when exposed
- The life time risk of developing genital warts & HPV infection is 10 & 50% respectively
- **Transmission of genital HPV infection**
 - Sexual contact
 - most common
 - partner with clinical/subclinical infection
 - Perinatal
 - may be transmitted
 - leads to laryngeal papillomas
 - Other means
 - digital lesions
 - usually in children
 - fomites
 - no good evidence
 -
- **Symptoms:**
 - Majority are asymptomatic
 - Little physical discomfort, but are disfiguring and psychologically distressing
 - Sometimes associated with itching, burning, bleeding, vaginal or urethral discharge, dyspareunia and outlet obstruction if lesion is large

Findings

a) Site

- Multifocal infection of anogenital skin lesions
- Most common at areas traumatized during sexual intercourse, but may occur at any site
- Urethral meatus is affected in 20-25% of males and 4-8% of females
- Anal warts seldom found proximal of the dentate line
- Intraanal warts more common with receptive anal intercourse
- Extragenital lesions may be seen on oral cavity, larynx, conjunctivae, and nasal cavity

b) Morphology

Clinical disease (Visible warts)

- Single or multiple lesions
- May coalesce into large plaques (more common in immunosuppressed and in diabetics)
- Colour varies from pinkish-raspberry to salmon-red (non-keratinised warts), grayish-white (heavily keratinized lesions), and ashen-grey to brownish-black (pigmented lesions)
- On warm, moist, non-hair bearing skin tend to be soft and non-keratinised
- On dry hairy skin are firm and keratinized

'Giant condyloma' (Buschke-Lowenstein tumour)

- Very rare variant of HPV 6 and 11-associated disease
- Characterized by increase in size and appears as large cauliflower-like masses
- Complex histological pattern with areas of benign condyloma intermixed with foci of atypical epithelial cells or well differentiated squamous-cell carcinoma
- Diagnosis requires multiple biopsies

Investigations

Diagnosis

- Naked eye examination in most cases
- If in doubt, or lesion is atypical or pigmented, biopsy for histology should be done
- **Assessment of lesions**
 - Examine external anogenital & surrounding skin under good illumination

- All women should have a speculum examination to identify presence of coexisting vaginal/or cervical warts
- Refer to gynaecologist for colposcopy for cervical lesions
- Refer to urologist for proximal urethral lesions
- Anoscopy/proctoscopy up to dentate line if anal warts present or history of anal receptive sex

Treatment

Natural course of HPV infection

Untreated visible warts may resolve spontaneously (40%), remain the same, or increase in size

Primary goal

Removal of visible / symptomatic warts

Indications for treating genital warts

- Cosmetic reasons
- Reduce transmissibility (no controlled studies)
- Provide relief from symptoms
- Improve self esteem

Effects of treatment

- May reduce HPV DNA in genital tissue, and therefore reduce infectivity
- Does not eradicate HPV or reduce the risk of genital cancer
- No controlled studies on effects of treatment and HPV transmission rates
- Recurrence rates often 20—30% or more

Treatment

- All treatments have significant failure and relapse rates
- Treatment decisions should be made after discussing the appropriate options with the patient, taking into account their preference and convenience
- Treatment may involve discomfort and local skin reactions. information on management of treatment side effects is recommended
- Local anaesthetic creams with or without 2% lignocaine injection . could be used before ablative therapy to minimize discomfort.
- Adrenaline-containing anaesthetic should be avoided on the penis and around the clitoris (due to vasoconstriction)

Treatment choices

- The evidence to guide towards first and second line treatments is not strong
- Treatment choice depends on a number of factors including the number, size, site, morphology, patient preference, cost, convenience, adverse effects, pregnancy, and provider experience
- Soft non-keratinised warts respond well to podophyllin, cryotherapy, excision or electrosurgery
- Keratinised lesions are better treated with physical ablative methods such as cryotherapy, excision or electrocautery
- Imiquimod may be suitable for both types
- People with a small number of low volume warts irrespective of type are best treated with ablative therapy from the outset

Treatments Available for External Genital & Perianal Warts

Patient Applied	Provider Administered
– Podophyllin 10–25% 1–2 times per week – Imiquimod 5% cream 3 times per week up to 16 weeks	– Cryotherapy with liquid nitrogen cryoprobe once in 1–2 weeks – Trichloroacetic acid 40–100% weekly – Scissor excision – Surgery – Electro surgery – Laser surgery

Patient applied therapies

Can be safely applied by the patient (refer to appropriate section)

Provider administered therapies

Initial treatment is usually with locally applied caustic agents.

It is usual to start with podophyllin – a cytotoxic agent – which should be applied to the lesions in strengths of 10% or 25% in industrial spirit and repeated weekly (refer appropriate section).

If it is ineffective after two or three weeks, the more caustic agent, glacial TCA 50-100% may be used. This agent is more useful for hyperkeratotic warts but even so these warts are often resistant and cryotherapy or electro surgery will be needed.

Electrosurgery, or surgical excision should be considered at an earlier stage if the warts are large or numerous.

General Guide to the Treatment of Genital Warts

Site or type of warts	Start with	One week	Two weeks	Three weeks	Four weeks
Few, small, and soft	10-25% podophyllin or cryotherapy	→	→	Trichloroacetic acid	Cryotherapy, electrocautery
Solitary, large, and discrete	Electrocautery diathermy, excision or cryotherapy				
Extensive, multiple vegetations	10-25% podophyllin or or trichloroacetic acid	→		→	Cryotherapy, surgical excision
Hyperkeratotic or keratinised	Trichloroacetic acid or cryotherapy	→		Electrocautery, diathermy	
Intrameatal	Cryotherapy	→		Electrocautery, cryotherapy	
Cervical	Colposcopy ± biopsy → cryotherapy, laser				
Vaginal	Cryotherapy or trichloroacetic acid				
Perianal	Cryotherapy				
Pregnancy	None – unless discrete small vaginal, vulval, or introital, then use trichloroacetic acid or cryotherapy				* Note : do not use podophyllotoxin, podophyllin, fluorouracil, or imiquimod

** Imiquimod works well for extensive multiple warts and recurrent warts*

Overview of Clearance and Recurrence Rates with Different Treatments for External Genital Warts

Patient Applied Therapies			
Treatment	Clearance Rate (%)		Recurrence Rate (%)
	End of treatment	≥ 3 months	
Imiquimod	50 – 62	50 – 62	13 – 19

Provider Applied Therapies			
Treatment	Clearance Rate (%)		Recurrence Rate (%)
	End of treatment	≥ 3 months	
Podophyllin	32 – 79	22 – 73	11 – 65
TCA	50 – 81	70	36
Surgical Excision	89 – 93	36	0 – 29
Electrotherapy	93 – 94	78 – 91	24
Laser Therapy	27 – 89	39 – 86	<7 – 45
Cryotherapy	63 – 88	63 – 92	0 – 39
Interferon	19 – 62	36 – 62	0 – 33

Chemical Treatment – Patient Applied

Imiquimod (Aldara)

- Topical cell mediated immune response modifier – produces a TH1-like response when applied to skin infected with HPV
- Indication : Suitable for all external anogenital warts
- Method
 - Apply thin layer of 5% cream to external visible warts, then rub in until cream vanishes; apply 3 times weekly & wash off 6-10 hours later for up to 16 weeks
 - Response to treatment may be delayed for some weeks
- Cure rates 77% in women, 40% circumcised & 62% uncircumcised men over 4-12 weeks & low relapse rate
- Caution: Not approved for use in pregnancy / or internal vaginal/anal warts

Chemical Treatment – Physician Applied

Podophyllin resin

- Non-standardised cytotoxic compound
- Antiproliferative with cytotoxic, antimitotic action causing tissue necrosis
- Animal experiments indicate teratogenic and oncogenic properties but evidence of these in humans are lacking
- Indications : few, small and soft anogenital warts; Method
 - Use 10 –25% Podophyllin in compound tincture of Benzoin
 - (10% Podophyllin in compound tincture of Benzoin for vaginal and anal warts)
 - Apply carefully to the warts while avoiding surrounding normal tissue
 - Protect surrounding skin with vaseline
 - Allow treated area to dry before contact with normal tissue or mucosa, especially in anal warts. For vaginal warts, treated area must be dry before removing speculum
 - Instruct patient to wash it off thoroughly in 4-5 hours
 - Treat weekly
 - If poor response after 4 – 6 weeks of treatment, alternative treatment is indicated
- Caution
 - Limit application to 10cm² or 0.5ml for external warts and less than 2cm² for vaginal warts
 - Can cause serious systemic side effects if applied in excess
 - Avoid on cervix, vagina or anal canal, & in pregnancy - potentially teratogenic & oncogenic; (squamocolumnar junction prone to dysplasia) and increased systemic absorption likely if used internally
- Advise patients to use condoms
- Routine pap smear screening for women with anogenital warts
- Atypical or persistent warts should be biopsied

Trichloroacetic acid (TCA)

- Acts as caustic agent resulting in protein coagulation with cellular necrosis of wart tissue (destructive action)
- Indication : Few, small soft or hyperkeratotic warts; Used in most anatomical areas
- Method
 - Weekly application of TCA 40-100% solution in a specialist clinic setting only; small amount applied & allowed to dry until a white frosting develops
 - Intense burning sensation may be experienced for 5-10mins after application
- Caution
 - Ulceration into dermis may occur, therefore not recommended for large volume warts;
 - Wash excess with liquid soap or sodium bicarbonate (NaHCO₃)
 - Extremely corrosive; protect normal skin with vaseline

Physical Ablation

Excision

- Indication : Pedunculated warts, & small numbers of keratinised ones at anatomically accessible sites
- Best done by a specialist & requires local anaesthetic, because may cause scarring
- Haemostasis achieved with electrosurgery or use of a haemostatic agent
- Repeated as required

Cryotherapy

- Indication : Small to moderate number of warts
- Apply with liquid nitrogen spray, cryoprobe or cotton-tip applicator
- Causes cytolysis at dermo-epidermal junction resulting in necrosis
- Treatment applied until a “halo” of freezing seen a few mm around the lesion (about 10-20 seconds)
- A freeze, thaw, freeze technique should be used
- Local anesthetic (topical or injection) may facilitate treatment
- Randomised trials show similar response as compared to podophyllin, TCA, & cautery

Electro surgery

- Indication : Used in most anatomical areas

Laser surgery

- Indication: CO2 laser suitable for large volume warts & in difficult anatomical areas, e.g. urethral meatus or intra-anal, or extensive vaginal.
- May be useful for HIV-infected patients with very large external genital warts All electrosurgical & laser techniques result in a plume of smoke which contain HPV DNA & may cause infection of the respiratory tract in operating personnel. Therefore, masks should be worn & adequate smoke evacuation provided

Adverse Effects of Treatment

	Adverse Effects & Incidence (%)
Imiquimod	Erythema (70%); irritation, ulceration & pain (10%); burning, erosion, flaking, edema, induration, & pigmentary changes; minimal systemic absorption
Podophyllin	Local irritation, erythema, burning, & soreness (75%); possible mutagenicity, oncogenecity Application of large quantities may cause peripheral neuropathy, coma, and hypokalemia
Trichloroacetic Acid	Local pain & irritation; no systemic side effects
Cryotherapy	Pain or blisters at application site (20%)
Surgical Excision	Pain (100%), bleeding (40%), scarring (10%), risk for burning and allergic reaction from local anesthetic
Laser Treatment	Similar to surgical excision; risk for HPV via smoke plume

Advice

- Detailed explanation of condition with long term implications for their health and their partners
- Reinforced with clear and accurate written information
- Condom usage with regular sex partners not shown to affect treatment outcome, although one study suggested a reduction in new HPV associated lesion formation
- Use of condoms may prevent transmission of HPV to uninfected partners and should be encouraged
- Routine (every 3 years) pap smear screening for women with anogenital warts

Contact Tracing

- Encourage patient to bring their current partner/s
- Current sexual partner may benefit from assessment as
 - they may have undetected genital warts
 - undetected other STIs
 - need explanation and advice about disease process in partner
- Tracing of previous sexual partner is not recommended

Prevention

Primary Prevention

HPV vaccines

- Quadrivalent vaccine (National HPV vaccination program for all girls 13 year old)
- Bivalent vaccine
- Both vaccines have a favourable tolerability profile.
- The quadrivalent vaccine has been shown to be highly effective in preventing cervical cancer, cervical intraepithelial neoplasia (CIN) 2/3, adenocarcinoma in situ (AIS), vulva and vaginal dysplasia, and other anogenital diseases, including genital warts caused by HPV 6, 11, 16, and 18 in women naïve to the relevant HPV types. This vaccine is recommended.
- The bivalent vaccine has been shown to be highly effective in preventing cervical cancer, CIN 2/3 and AIS caused by HPV 16 and 18 in women naïve to the relevant HPV types but not against genital warts caused by HPV 6 and 11.

Special Considerations

I) Anatomical Sites

Intravaginal

- Cryotherapy, TCA, & electrosurgery are recommended treatments

Peri- / intra-anal warts

- Treatment options include TCA, cryotherapy, electrosurgery, , and laser

Cervix

- Best practice suggests colposcopy & biopsy before treatment to exclude CIN & cervical cancer
- If no CIN, cryotherapy, electrosurgery or TCA may be used
- If CIN present, refer to gynaecologist for further treatment

Urethral lesion

- If base of lesion seen, treat with cryotherapy orelectrosurgery
- Lesions deeper in urethra should be referred to Urology for surgical ablation under direct vision

Large warts

- Warts > 10mm in diameter
 - Surgical excision as primary therapy, cautery or cryotherapy

- Alternatively, imiquimod cream applied for 3-4 treatment cycles may reduce size of warts & improve surgical outcome
- If > 50% reduction in size, continue until warts clear or until 8 cycles
- If < 50% reduction in size, surgical ablation or other ablative therapy initiated

II) Pregnancy

Warts may increase in size due to altered immunity and increased blood supply

- Aim to minimize number of lesions at delivery to reduce neonatal exposure to the virus
- Potential problem for children are development of laryngeal papillomatosis and anogenital warts. Mode of delivery may be discussed with obstetrician. Avoid podophyllin, & 5 FU because of possible teratogenic effects
- Imiquimod not approved for use in pregnancy
- During pregnancy, treatment is usually not recommended; if necessary, use cryotherapy, TCA, excision, laser/cautery
- Very rarely a caesarean section is indicated because of blockage of vaginal outlet with warts or presence of gross cervical warts

III) Immunosuppressed

- Patients with impaired cell mediated immunity (CMI) – e.g. renal transplant, HIV infection may present with larger and more difficult to treat warts
- Response to treatment may be poorer with increase in relapse rate, and higher risk of dysplasia
- Careful follow-up is required to exclude any transformation to malignancy

IV) Cervical cytology (PAP smear) for women with anogenital warts

- No change to screening intervals
- Follow guidelines for abnormal smears as defined

Follow Up

- Review at the end of treatment (refer table) to monitor response and assess need for change in therapy
- If original lesion responds but new lesions evolve, continue with current regime
- Change indicated if
 - patient not tolerating current therapy
 - under 50% response to current treatment (after 3 treatment cycles)

TRICHOMONIASIS

Aetiology

Trichomonas vaginalis

Incubation Period: 4 days to 4 weeks

Presentation and Findings:

Female:

- Diffuse, malodorous, frothy, yellow-green vaginal discharge with vulval itching ± dyspareunia, ± dysuria, vulvo-vaginal soreness (vulvitis and vaginitis)
- Strawberry cervix (punctate cervical erythema)
- Occasionally lower abdominal discomfort
- 10-50% asymptomatic

Male:

- Usually contacts of infected women
- Scanty or moderate urethral discharge (50-60%)
- Rarely prostatitis or balanoposthitis
- 15-50% asymptomatic

Complications:

- Premature rupture of membranes
- Low birth weight
- Preterm delivery

Investigations:

Female:

1. Saline wet smear from posterior fornix

- microscopy should be performed as soon as possible after the sample is taken as motility diminishes with time
- motile flagellates, oval or pear shaped organism with jerky movement,
- positive in 40-80% of cases

2. Cervical pap smear

- sensitivity 60%, but high rate of false positives;
- not recommended

3. PCR

- sensitivity and specificity almost 100%

Male:

1. Saline wet smear

- positive in 30% of cases

2. PCR

- sensitivity and specificity almost 100%

Treatment

Recommended (A, Ia):

- Metronidazole 400 mg b.d. P.O. for 5 – 7 days; OR
- Metronidazole 2 g single dose P.O. (pregnancy category B); OR
- Tinidazole 2 g single dose P.O (pregnancy category C)

(95% cure rate, resistance rarely reported, published data suggest no increased teratogenic risk in pregnancy with Metronidazole)

Pregnancy

- High dose of Metronidazole or Tinidazole is not recommended in pregnancy and breastfeeding (due to metallic taste in breast milk)

Treatment failure with Metronidazole (Metronidazole Resistant Trichomoniasis)

- Higher dose of Metronidazole
 - 2gm daily for 3-5 days
 - 400mg t.d.s orally with 1 g PR daily for 7 days.
- Higher dose of Tinidazole 2gm b.d for 2 weeks with or without intra-vaginal Tinidazole
- Clotrimazole intra-vaginal cream

Advice

- No sexual intercourse until one week treatment completed
- No alcohol (antabuse effect with metronidazole)

Contact Tracing

- Encourage patient to bring their current sexual partners for examination and investigation for full range of STIs
- Treat sex partners epidemiologically

Follow-Up

- Repeat Saline wet smear in 7 – 10 days

ANOGENITAL CANDIDIASIS**Aetiology**

Candida albicans 80 – 92%

Non-albicans species e.g. *C.glabrata*

Presentation:**Female:**

- Vulval itching & soreness
- Thick white curdy vaginal discharge that is worse before menses ± dyspareunia and ± dysuria

Male:

- Rash on glans penis
- Penile soreness/itch

Findings**Female:**

- Vulval erythema with satellite lesions
- Fissures at perineum
- Thick white curdy vaginal discharge

Male:

- Erythema over glans penis
- Fissures over prepuce
- Oedema of prepuce
- Phimosi

Investigations

- Lateral wall of vagina/subpreputial smear
 - 10% KOH microscopy (sensitivity 70%)
 - Gram stain for Gram-positive yeast-like cells (sensitivity 65 – 68%)
- Culture on Saboraud's agar

Treatment

Topical and oral azole therapies (Cure rate 80 – 95%)

Recommended (A, II):*Topical Therapy*

- Clotrimazole pessaries 200 mg daily for 3 nights; OR
- Clotrimazole pessaries 500 mg single dose

Alternative:*Topical Therapy*

- Nystatin pessaries 100,000 units o.n. for 2 weeks; OR
- Tioconazole pessaries 200 mg daily for 3 days; OR
- Tioconazole 6.5% ointment 5 g intravaginally-single application

Oral Therapy

- Fluconazole 150 mg single dose P.O.(pregnancy category C); OR
- Itraconazole 200 mg b.d. P.O. for 1 day (pregnancy category C)

Pregnancy:

- Topical therapies recommended for a longer period.
- Oral therapy is contraindicated

Advice

- Avoid local irritants e.g. perfumes, soap
- Avoid tight fitting synthetic clothing

Contact Tracing

No evidence to support epidemiological treatment of asymptomatic male sexual partners

Follow-Up

- 7 – 14 days after completion of therapy
- Repeat vaginal smear for candida

RECURRENT VAGINAL CANDIDIASIS

- 4 or more episodes of symptomatic candidiasis annually
- Predisposing factors: diabetes mellitus, HIV, corticosteroid use, frequent broad spectrum antibiotic use, immunosuppression

Treatment

Recommended regimen

- Fluconazole 150 mg P.O. weekly for 6 months; OR
- Clotrimazole vaginal pessary 500 mg weekly for 6 months; OR
- Itraconazole 400 mg P.O. monthly for 6 months

BACTERIAL VAGINOSIS / ANAEROBIC VAGINOSIS

Aetiology

- Bacterial Vaginosis (BV) is a polymicrobial clinical syndrome resulting from replacement of the normal hydrogen peroxide – producing *Lactobacillus* sp. in the vagina with high concentrations of anaerobic bacteria (e.g., *Prevotella* sp. and *Mobiluncus* sp. , *Gardnerella vaginalis*, and *Mycoplasma hominis*)
- Is not a STI but frequently detected during STI screening

Presentation

- Abnormal vaginal discharge with a strong fish-like odour, especially after sexual intercourse
- Dysuria and / or itching around the outside of the vagina
- 50% of women with BV are asymptomatic

Findings

- Discharge, if present, is usually homogeneous thin, white or grey, uniformly adherent – coating the walls of the vagina and vestibules
- Inflammation of the vaginal wall is absent

Investigations

Using the Amsel's criteria, diagnosis is made by the presence of any 3 out of the 4 features given below:-

1. Homogeneous, thin, white discharge that smoothly coats the vaginal walls
2. Presence of clue cells on microscopic examination
3. pH of vaginal fluid >4.5
4. Fishy odor of vaginal discharge before or after addition of 10% KOH (the Amine' Sniff test)

An alternative is to use a Gram stained vaginal smear, with the Hay/Ison criteria or the Nugent criteria. The Hay/Ison criteria is defined as follows:

- Grade 1 (Normal) : *Lactobacillus* morphotypes predominate
- Grade 2 (Intermediate) : Mixed flora with some Lactobacilli present, but *Gardnerella* or *Mobiluncus* morphotypes also present
- Grade 3 (Bacterial Vaginosis) : Predominantly *Gardnerella* and/or *Mobiluncus*

morphotypes. Few or absent *Lactobacilli*

The Nugent score is derived from estimating the relative proportions of *Lactobacilli* and bacterial vaginosis morphotypes to give a score of between 0 and 10. A score of < 4 is normal, 4 to 6 intermediate, and > 6 indicates bacterial vaginosis.

Treatment

Treatment is indicated for:

- Symptomatic women (Grade A)
- Women undergoing gynecological procedures (Grade A)
- Pregnant women

Recommended:

- Metronidazole 400 mg b.d. P.O. for 5 to 7 days (Grade A); OR
- Metronidazole 2 g single dose P.O. (Grade A)

Alternative:

- Intravaginal metronidazole 0.75% gel once daily for 5 days (Grade A); OR
- Intravaginal clindamycin 2% cream once daily for 7 days (Grade A); OR
- Clindamycin 300 mg b.d. P.O. for 7 days (Grade A)

Avoid alcohol intake with Metronidazole. It causes severe giddiness and nausea when taken together with alcohol (antabuse effect)

Allergy:

- Use 2% clindamycin cream for metronidazole allergic women.

Pregnancy and Breastfeeding:

BV during pregnancy is associated with adverse pregnancy outcomes:

- Premature rupture of the membranes
- Preterm labour
- Preterm birth
- Chorioamnionitis
- Postpartum endometritis

Meta-analyses have concluded that there is no evidence of teratogenicity from the use of metronidazole in women during the first trimester of pregnancy.

Metronidazole is excreted in breast milk and gives a metallic taste. Avoid high dose of metronidazole (2 gm single dose) if breast feeding.

Advice

Patients should be advised to avoid vaginal douching. Avoid use of shower gel, antiseptic soap and shampoos in baths (Grade C).

Contact Tracing

Routine screening and treatment of male partners are not indicated.

Follow - Up

- A test of cure is not required if symptoms resolve
- If treatment is prescribed in pregnancy to reduce the risk of preterm birth, a repeat test should be made after 1 month and further treatment is offered if there is evidence of recurrence of bacterial vaginosis

GRANULOMA INGUINALE (DONOVANOSIS)**Aetiology**

Klebsiella (Calymmatobacterium) granulomatis

Endemic in Western Guinea, South America, the Caribbean, Southeast India and aboriginal population in Australia

Incubation Period

1-4 weeks (up to 6 months)

Presentation / Findings

Present initially as a single or multiple painless nodules or papules which erode to form beefy exuberant, granulomatous ulcers with rolled and elevated border, usually painless, bleeds easily and enlarge slowly. It may auto-innoculate to form kissing lesions or form pseudobuboes in the inguinal area.

It may also present as hypertrophic or necrotic lesions.

Complications include haemorrhage, genital lymphoedema, genital mutilation and cicatrization. Squamous cell carcinoma may complicate the presentation.

Investigations

1. Tissue smears to look for bipolar staining rod-like bacilli resembling safety pins (Donovan's bodies) with Wright or Giemsa stain. Silver stains have a high sensitivity (60-80%)
2. Tissue biopsy
3. PCR studies from tissue
4. Tests to exclude other causes of genital ulcers

Treatment**Recommended Regimen**

- Azithromycin 1gm orally once a week for 4 weeks or 500 mg daily for 7 days (Grade B, Ib)

Alternative Regimen

- Cotrimoxazole 2 tabs b.d. P.O. for 2-3 weeks (Grade B, Ib); OR
- Ceftriaxone 1 g daily I.M. or I.V. for 2-3 weeks (Grade B, Ib); OR
- Erythromycin 500 mg q.i.d. P.O. for 2-3 weeks; OR
- Doxycycline 100 mg b.d. P.O. for 2-3 weeks (Grade C, IV)

Gentamycin 1mg/kg 8hourly I.M. or I.V. is used as an adjunct treatment if the above regimens do not give a satisfactory response after a few days

Treatment should be continued until lesions have healed

Pregnancy

Erythromycin is recommended.

Infants born to mothers with untreated genital lesions are at risk of infection and a course of prophylactic antibiotics should be considered.

Contact Tracing

Sexual partners must be screened and treated

Follow-up

Weekly till ulcers have healed.

CHANCROID

Aetiology

Haemophilus ducreyi

Incubation Period

3 to 10 days

Presentation / Findings

Divided into those at the site of primary inoculation and at the regional lymph nodes

I) At the site of primary inoculation

In males, ulcers are found on the prepuce near the frenulum or in the coronal sulcus. In females ulcers are found at the entrance to the vagina, particularly the fourchette.

The ulcers are classically described as:

- Multiple, painful
- Not indurated (“soft sore”)
- With a necrotic base and purulent exudate
- Bordered by ragged undermined edges
- Bleeds easily on contact

II) At the regional lymph nodes

Painful unilateral inguinal adenitis is a characteristic feature (50%), leading to the formation of buboes (purulent abscess of the inguinal lymph nodes)

Buboes are fluctuant and may rupture, releasing thick pus, and may result in extensive ulceration.

A probable diagnosis of chancroid can be made if all the following criteria are met:-

1. The patient has one or more painful genital ulcers
2. The patient has no evidence of *T.pallidum* infection by dark field examination ulcer exudates or by serologic testing for syphilis performed at least 7 days after the onset of ulcers
3. The clinical presentation, the appearance of genital ulcers, and if present, the presence of regional lymphadenopathy are typical for chancroid

4. Test results for herpes simplex virus performed on the ulcer exudates are negative

Investigations

Specimen collection

- Genital specimens for *H.ducreyi* should be collected from the base and the (undermined margins of the chancroid lesion after cleansing the area by flushing with sterile physiological saline.
- Collect the material from the ulcer with calcium alginate, Dacron or cotton swabs
- Direct plating on selective media is preferable as it results in the highest recovery. Examples of media that have been shown to give optimal recovery are gonococcal agar supplemented with 2% bovine hemoglobin and 5% fetal calf serum (containing 1% IsovitaleX supplement and 3µg/mL of vancomycin) and Mueller-Hinton agar supplemented with 5% chocolate horse blood (containing 1% IsovitaleX supplement and 3µg/mL of vancomycin)
- Specimen plated directly in the clinic or sent within 4 hours to the laboratory using Amies transport media.

1. Gram stain of scrapings from the ulcer base or pus aspirated from the bubo

- Gram negative coccobacilli, with characteristic appearance (“school of fish”) – low sensitivity; sensitivity 50-60%, specificity 50-100%

2. Culture

- Special culture medium for Chancroid – refer Specimen collection
- Sensitivity 80%

3. PCR (> 95% sensitivity)

4. Screening for other possible causes of genital ulcers:

- *Treponema pallidum*
- Genital herpes
- Lymphogranuloma venereum (LGV)
- Donovanosis
- HIV
- Neoplasia

Treatment

Recommended:

- Azithromycin 1 g single dose P.O (Grade A, Ib); OR
- Ceftriaxone 250 mg single dose I.M. (Grade A, Ib); OR

- Ciprofloxacin 500 mg b.d. P.O. for 3 days (Grade B, Ib); OR
- Erythromycin base 500 mg q.i.d. P.O. for 7 days (Grade B, Ib)

Alternative

- Spectinomycin 2 g single dose I.M (Grade B, IIa); OR
- Norfloxacin 800 mg single dose P.O (Grade B, Ib)

Epidemiological Treatment:

Sexual contacts within 10 days before onset of the patient's symptoms should be examined and treated even in the absence of symptoms, as asymptomatic carriage of *H. ducreyi* is not uncommon. (Grade C, IV)

Special Considerations

A. Management of fluctuant buboes

- Needle-aspirate fluctuant buboes from adjacent healthy skin is simpler and safer than incision, which is prone to sinus formation.
- Aspirate from top of bubo not from the base

B. Treatment of pregnant or lactating mothers and children

- The safety of azithromycin for pregnant and lactating women has not been established.
- Ciprofloxacin is contraindicated for pregnant and lactating women, children, and adolescents less than 18 years of age.
- The erythromycin or ceftriaxone regimens should be used.
- No adverse effects of chancroid on pregnancy outcome or on the fetus have been reported.

C. HIV patients

- Healing of ulcer may be slower with a high rate of treatment failure
- Ceftriaxone and azithromycin regimens should be used among persons known to be infected with HIV only if follow-up can be assured

Advice

- Patients should be advised to avoid unprotected sexual intercourse until they and their partner(s) have completed treatment and follow-up.
- Patients and partner(s) should be given a detailed explanation of the condition with particular emphasis on the long-term implications to their health. This should be reinforced by giving them clear and accurate written information.

Follow-Up

- Patients should be re-examined 3-7 days after initiation of therapy.
- If treatment is successful, ulcers improve symptomatically within 3 days and substantial re-epithelization occurs within 7 days after onset of therapy.
- The time required for complete healing is related to the size of the ulcer (and perhaps HIV); large ulcers may require more than 2 weeks.
- Treatment failures:
 - Investigate for possible co-infections with *T. pallidum* or HSV
 - Determine possible resistance by isolation of *H. ducreyi* and susceptibility testing
 - Clinical resolution of fluctuant lymphadenopathy is slower than that of ulcers and may require frequent needle aspiration (or drainage)

Complications (mostly in men)

- Phimosis
- Phagedema ulcers

LYMPHOGRANULOMA VENEREUM [LGV]

Lymph granuloma venereum [LGV] is on the rise in the Western Europe, the USA and Australia; it is restricted mainly MSM communities and those with HIV. Of late it has appeared in the heterosexual communities in Spain and Portugal. It is endemic in Africa and India, being responsible for 2 to 10% of GUD.

Aetiology

- It is caused by *C. trachomatis* serovars L1 –L3.
- The L2b serovar has been responsible for the lesions in the MSM communities (men who have sex with men).
- Most have associated HIV infection.

Incubation Period: 5 to 30 days

Clinical Features

Three separate stages seen:

1) Primary Lesion:

- After a transient interval a small painless papule or pustules is present; it may erode forming herpetiform ulcer healing within a week.
- Urethral, vaginal, cervical and rectal: muco-purulent discharge occasionally seen.

2) Secondary Stage:

- 2 - 6 weeks – the “*inguinal stage*.” The inguinal and/or the femoral lymph nodes get enlarged; typically unilateral in 2/3rd of cases.
- Typically unilateral inflamed suppurating abscesses, known as ‘buboes’ rupturing in a third of cases.
- The typical ‘grooves sign’ is seen in 15 - 20% cases only; Intra-abdominal or retroperitoneal lymphadenopathy may occur causing low back pain and/or abdominal pain.
- Constitutional symptoms and signs: At times low grade pyrexia, arthralgia, myalgia, malaise, arthritis, pneumonitis, peri-hepatitis (FHCS) maybe present.

3) Tertiary stage:

- The ‘*ano-genito-rectal syndrome*’ present mostly in women.

- They develop proctocolitis followed by peri-rectal abscesses, mimicking Crohn's disease.
- Complications include fistulae, strictures leading to stenosis of rectum and and lymphoedema of the genitalia (elephantiasis)

Epidemic amongst the MSM

- Proctitis is the main presenting symptoms
- Haemopurulent discharge, anorectal pain, tenesmus, rectal bleeding and constipation are common presenting features.
- Examination: shows hemorrhagic proctitis, purulent edema, and distal granular proctitis.

Investigations and Diagnosis

Diagnosis is based on clinical suspicion, epidemiological information, and the exclusion of other etiologies of genital or rectal ulcers, inguinal lymphadenopathy, or proctocolitis. Special attention should be given to MSM's and HIV patients.

- a) PCR (Grade B, III)
- b) If above are not available, serology for Chlamydia genus specific assay for a presumptive diagnosis. A high titre IgA anti-MOMP antibodies suggestive of LGV; however, low titre does not rule out LGV. (Grade B, III)
- c) Confirmation of LGV by detecting serovar-specific *C. trachomatis* DNA

Treatment

Complete treatment of patients with LGV includes appropriate antimicrobial coverage and drainage of infected buboes.

The recommended medical treatment for LGV involves one of the following antibiotic regimens:

- Doxycycline 100 mg PO b.d for 21 days (Grade B, IIb)
- Erythromycin base 500 mg PO 6 hrly for 21 days (Grade B, III)
- Azithromycin 1 gm weekly for 3 weeks (Grade C, IV)

Epidemiological Treatment

Sex partner/s that have had contact with the patient within the past 90 days should be evaluated and treated if symptomatic. If no symptoms are present, they should be treated for exposure as empirically as follows:

- Doxycycline 100 mg PO b.d for 7 days (Grade C, IV)

- Azythromycin 1 gm PO stat. (Grade C, IV)

Pregnancy

- Erythromycin base 500 mg PO 6 hrly for 21 days

Surgical Care

Needle aspiration or incision and drainage of involved inguinal nodes may be required for pain relief and prevention of ulcer formation. (Grade C, IV)

Some of the late complications of the third stage of LGV may require surgical repair. (Grade C, IV)

Advice

Patients should be given a detailed explanation of their condition and the long term implications to their health and their partner(s). Unprotected sexual intercourse should be avoided until the patients and their partner(s) have completed treatment and follow-up.

Follow-Up

Patients should be followed up until their signs and symptoms have resolved (within 3-6 weeks). Fibrotic lesions and fistulae may need reconstructive genital surgery.

MODIFIED SYNDROMIC APPROACH TO STI MANAGEMENT

WHO advocates syndromic approach for management of STI but due to better laboratory facilities, Malaysia encourages an aetiological approach to STI treatment. Modified Syndromic Approach may be considered in remote areas with poor access to laboratory facilities.

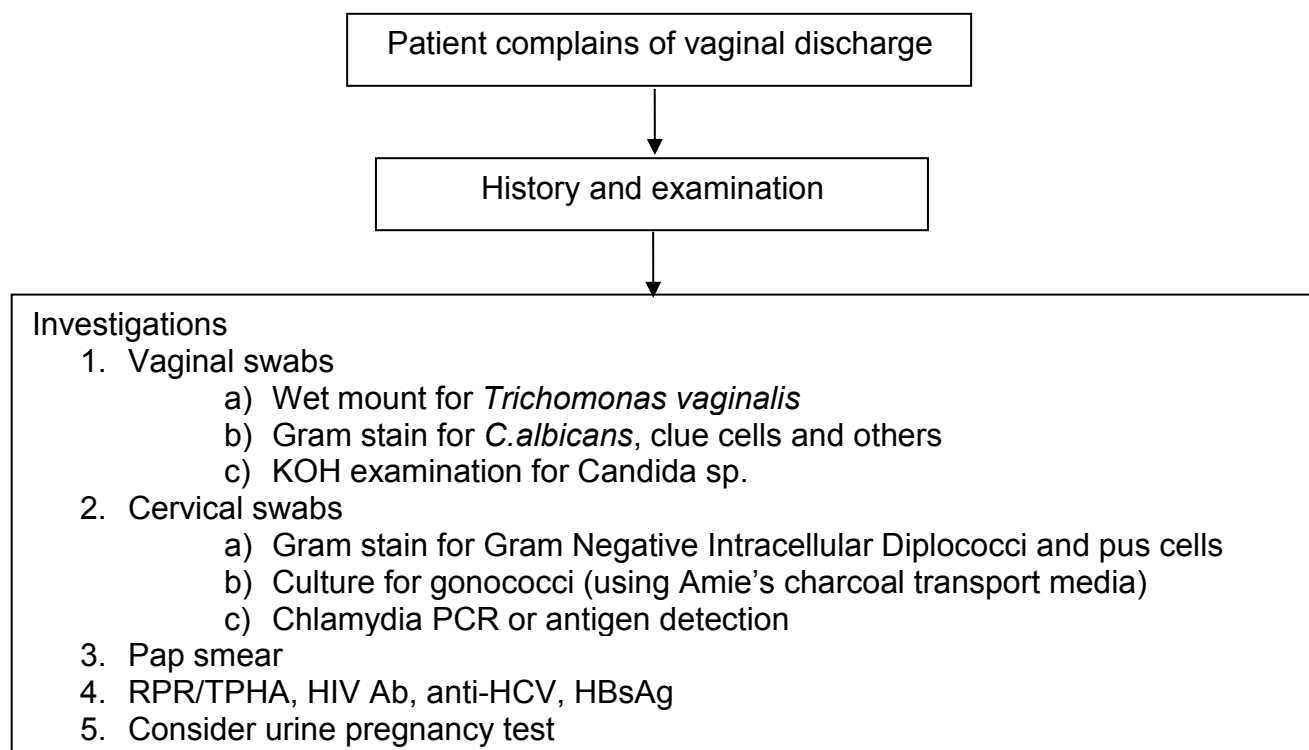
ABBREVIATIONS

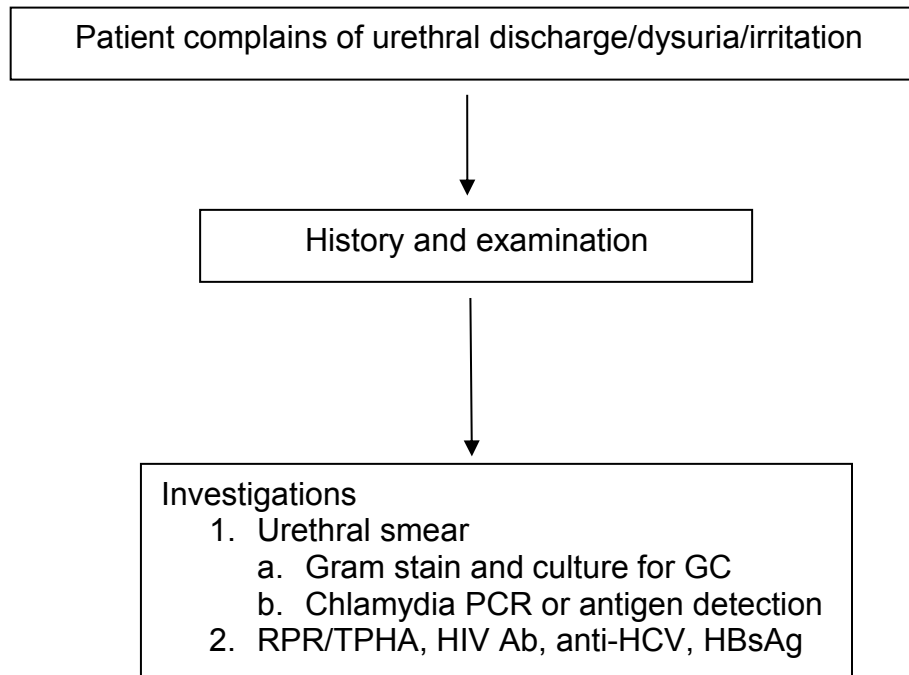
Ab	Antibody
Ag	Antigen
AIDS	Acquired Immunodeficiency Syndrome
Anti-HCV	Hepatitis C antibodies
BASHH	British Association for Sexual Health and HIV
BV	Bacterial Vaginosis
CDC	Centers for Disease Control and Prevention
CIN	Cervical Intraepithelial Neoplasia
CMI	Cell-mediated Immunity
CNS	Central Nervous System
CS	Caesarian Section
CSF	Cerebrospinal fluid
DIF	Direct Immunofluorescent Assay
DFA	Direct Fluorescent Antibody
DGI	Disseminated Gonococcal Infection
DNA	Deoxyribonucleic acid
EB	Elementary Bodies
EIA	Enzyme immunoassay
ELISA	Enzyme-linked immunosorbent assay
EPT	Expedited partner therapy
FBC	Full Blood Count
FDA	Food and Drug Administration
FTA-ABS	Fluorescent treponemal antibody absorption
GC	Gonococcus
GH	Genital Herpes
GU	Genito-urinary
HAART	Highly Active Antiretroviral Therapy
HBsAg	Hepatitis B surface Antigen
HBV	<i>Hepatitis B virus</i>
HCV	<i>Hepatitis C virus</i>
HIV	<i>Human immunodeficiency virus</i>
HPV	<i>Human papilloma virus</i>
HSV	<i>Herpes simplex virus</i>
IFA	Immunofluorescent assay
Ig	Immunoglobulin
I.M.	Intramuscular
IUD	Intrauterine device
I.V.	Intravenous
KOH	Potassium hydroxide
LGV	Lymphogranuloma venereum
MMWR	Morbidity and Mortality Weekly Report
MSM	Men who have sex with men
NAAT	Nucleic acid amplification test

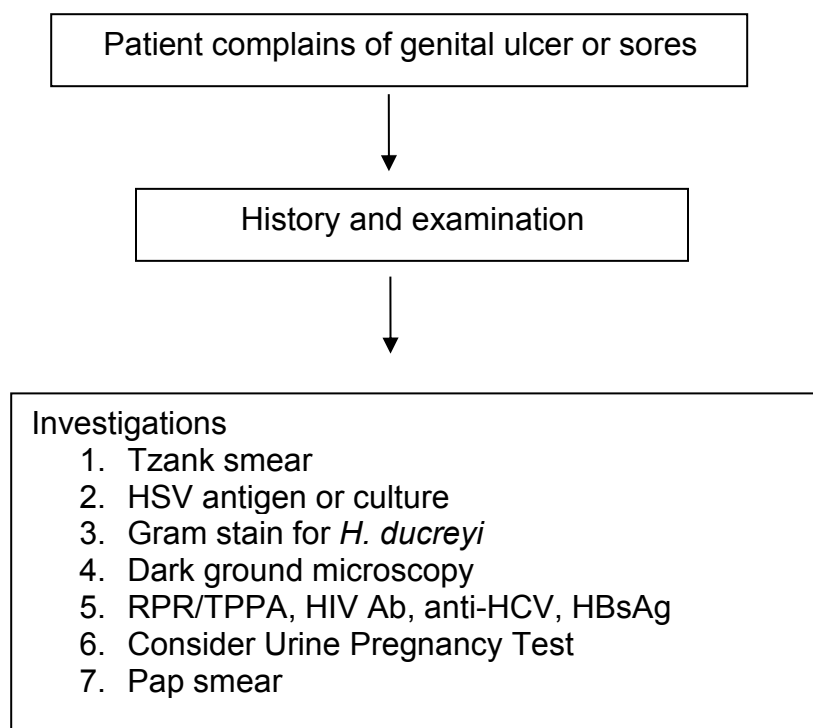
NGU	Nongonococcal urethritis
NSU	Nonspecific urethritis
NSGI	Nonspecific genital infection
Pap	Papanicolaou
PCR	Polymerase chain reaction
PID	Pelvic Inflammatory Disease
P.O.	Per oral/ by mouth
QOL	Quality of life
QRNG	Quinolone resistant <i>Neisseria gonorrhoeae</i>
RB	Reticulate bodies
RNA	Ribonucleic acid
RPR	Rapid Plasma Reagin
SARA	Sexually acquired reactive arthritis
STD	Sexually transmitted disease
STI	Sexually transmitted infection
TCA	Trichloroacetic acid
TPHA	<i>Treponema pallidum</i> haemagglutination
TPPA	<i>Treponema pallidum</i> particle agglutination
VDRL	Venereal Disease Research Laboratory
VZV	Varicella zoster virus
WBC	White blood cell
WHO	World Health Organization
WSW	Women who have sex with women

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6. Malaysia Global AIDS Response Report 2015

VAGINAL DISCHARGE

URETHRAL DISCHARGE

GENITAL ULCER

ORAL DESENSITISATION PROTOCOL

Dose	Penicillin V Suspension (units/ml)	Amount		Cumulative dose (units)
		ml	units	
1	1000	0.1	100	100
2	1000	0.2	200	300
3	1000	0.4	400	700
4	1000	0.8	800	1500
5	1000	1.6	1600	3100
6	1000	3.2	3200	6300
7	1000	6.4	6400	12 700
8	10 000	1.2	12 000	24 700
9	10 000	2.4	24 000	48 700
10	10 000	4.8	48 000	96 700
11	80 000	1.0	80 000	176 700
12	80 000	2.0	160 000	336 700
13	80 000	4.0	320 000	656 700
14	80 000	8.0	640 000	1 296 700

TABLE: LEVELS OF EVIDENCE AND GRADING OF RECOMMENDATIONS

The levels of evidence and recommendations have been graded as below

Levels of evidence

Levels of evidence	Explanation
Ia	Evidence obtained from meta analysis of randomized controlled trials (RCT)
Ib	Evidence obtained from at least 1 RCT
IIa	Evidence obtained from at least one well designed controlled without randomisation
IIb	Evidence obtained from at least one other type of well designed quasi experimental study
III	Evidence obtained from well designed descriptive studies
IV	Evidence obtained from expert committee reports or opinions and/or clinical experience of respected authorities

Grading of recommendations

A	Evidence at level Ia or Ib
B	Evidence at level IIa, IIb or III
C	Evidence at level IV

Adopted from BASSH Guidelines, 2010